



Incorporating novel input variable selection method for in the different water basins of Thailand

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ABSTRACT

Selecting appropriate input variables for developing a rainfall prediction model is significantly difficult. The present study proposed an innovative framework for input variable selection (IVS) model bootstrapped long short-term recurrent neural network (BTSP-LSTM-RNN) to identify relevant variables for monthly rainfall forecasting. Monthly meteorological and large-scale climatic variables (LCVs) from 1993 to 2022 at two selected river basins in the northern region of Thailand were used for model development. The proposed BTSP-LSTM-RNN model results were compared with the support vector regression with recursive feature elimination (SVR-RFE) and Gradient boosting (GB) by statistical metrics such as coefficient of determination (R^2), mean absolute error (MAE), relative root mean squared error (RRMSE), Pearson's correlation coefficient (r) and mean absolute percentage error (MAPE). BTSP-LSTM-RNN demonstrated exceptional performance, boasting a higher R^2 (0.84), MAE (92.28), RRMSE (10.36) in the Wang basin, and R^2 (0.83), MAE (242.60), RRMSE (9.93) in the Nan basin. BTSP-LSTM-RNN also achieved the lowest MAPE of 29.82%. Based on this IVS model results, two input variable combinations (IVCs) were designed. IVC-1 is based on BTSP-LSTM-RNN selection, and IVC-2 is an original set of variables. LSTM-RNN, multi-layer perceptron artificial neural network (MLP-ANN), and ensemble model with bootstrapping on random forest (RF) were employed for monthly prediction. When BTSP-LSTM-RNN's selected input variables from IVC-1 are utilized, the BSTP-RF model demonstrates robust performance. It achieves a high R^2 (0.82), a low RRMSE (10.13%) suggests accurate predictions, and the r of 0.91 further supports the model's strong linear relationship with observed rainfall data. Based on prediction model results, the BTSP-LSTM-RNN (IVS) model plays a pivotal role in the selection of input variables for rainfall forecasting and its impact on the performance of prediction models (BSTP-RF, MLP-ANN, and LSTM-RNN). These results consistently underscored the pivotal role of the BTSP-LSTM-RNN IVS model in enhancing the precision and reliability of rainfall predictions.

1. Introduction

Climate change adaptation is one of the most challenging tasks in managing water resources. Rainfall also causes many severe events, such as storms, resulting from climatic changes, which are becoming more common, although these events will trigger an assortment of disasters in the future [1]. Accurate rainfall forecast models can assist in water resource management and planning. The unpredictability and complexity of rainfall are the major factors limiting the effectiveness of rainfall forecasting [1,2]. Scientific studies have demonstrated rainfall

patterns' complexity due to their intricate relationship with natural variabilities encompassing randomness, non-stationarity, complexity, and non-linearity [3]. Rainfall forecasting encompasses both short-term and long-term perspectives, including immediate hourly forecasts for flood management and extended prognostications covering spans like the forthcoming six months. Long-term rainfall forecasts are applied in crucial operational domains, including irrigation and management of water resources, risk assessment of flooding, and mitigation or the planning of hydropower initiatives [4,5]. In recent years, a vast range of techniques for rainfall forecasting has been employed to explore the

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influence of meteorological and large-scale climatic variables on rainfall variability [6,7]. Two distinct methodological approaches, dynamical and empirical, have been utilized for rainfall forecasting [8]. A dynamical approach is harnessed within the paradigm to formulate and solve equations conducive to rainfall forecasting. In contrast, the empirical approach entails a retrospective analysis of previous oceanic and atmospheric data to cultivate a simulation model for forecasting rainfall during forthcoming periods [2,9,10]. This study is based on empirical approaches. A central dilemma in developing forecasting models revolves around thoroughly selecting input variables. This dilemma is fundamental to the query about the optimal number of input variables that assure inclusion within a given model. The process of selecting input variables, known as input variable selection (IVS), has consistently gained limited attention within the academic discourse. IVS methodologies are conventionally employed to discover the most efficacious input parameters from an expansive array of potential candidates. Here, efficiency is defined by the parameter's utmost pertinence to the desired output concurrently to minimize duplicative attributes amongst the set of prospective inputs [11]. Despite accumulating numerous hydro-meteorological studies, no authentic approach universally adapts rainfall forecasting phenomena across diverse conditions, particularly those characterized by distinct region attributes driven by specific physical mechanisms such as periodicity or stochasticity, remains elusive. Within this context, a universally endorsed model capable of consistently outperforming alternative models under disparate hydrological conditions has yet to emerge. Historical and climatic data's dynamic and non-stationary nature presents challenges in attaining cohesive predictions through multiple models. Consequently, the constraint develops for researchers to build more robust and efficient models, leveraging the available historical and climatic data. Numerous artificial intelligence-based approaches applications in hydrology did not depict a methodical IVS approach and depended on the techniques elaborated in previous investigations about the system, such as linear cross-correlation, mutual information scores, and trial-and-error [12]. Furthermore, some statistical approaches are used by many researchers, but all techniques have limitations. For example, the trial-and-error approach, in which an instinctive-force technique is employed to identify the optimal inputs, is computationally costly, particularly for ANN systems with various inputs. [3,13]. The most generally utilized IVS approach, linear cross-correlation, is reduced to distinguishing discrete, linear connections among the output or particular potential inputs. [11]. While cross-correlation methods might be beneficial in minimizing the input number comprising the set of variables by offering modelers with estimated time lags among observing points, they cannot capture the non-linearities, interdependencies, and redundancies of the typical hydrological systems. [14,15]. Similarly, the mutual information scores (MIS) feature selection method uses mutual information to select the most relevant input variables for rainfall forecasting. MIS measures the degree of dependence between two variables, and this method selects the input variables with the highest mutual information with the target variable (rainfall) [16]. Furthermore, Table 1 presents a compilation of diverse methodologies and outcomes across several studies in rainfall forecasting. Dariane, Ashrafi Gol, and Karami, (2019) used a Genetic Algorithm (GA) combined with the self-organized neural network (SOM) employing the winGamma software, which has showcased enhanced dependability for long-term rainfall forecasting. This method demonstrates its potential for reliable IVS, particularly in the Iranian context, where large-scale climatic signals such as the Pacific Decadal Oscillation (PDO), Trans-Niño Index (TNI), and SST-Niño 3 play a pivotal role in precise predictions [17]. Another investigation introduces the circle-inspired optimization algorithm-deep long short-term memory approach, displaying heightened performance levels in rainfall forecasting. With minimal relative absolute error (RAE) at 0.023 and mean square error (MSE) and root mean square error (RMSE) values of 0.151 and 0.389, respectively, this method showcases a promising avenue for forecast enhancement [1]. Additionally, a monthly rainfall

forecast model using deep neural networks (MRDNN) has been proposed, exhibiting an approximately 70% accuracy for a 3-month leading time of forecast. This model draws on large atmospheric variables (LAV) and a meteorological dataset to provide improved predictive capabilities [7]. Further exploration involves the application of ANN and the predictive ocean-atmosphere model for Australia (POAMA) for monthly rainfall forecasting in Queensland, Australia. This method exhibits superior performance, evidenced by reduced RMSE and mean absolute error (MAE) as well as elevated correlation coefficients (r), highlighting its advantage over the existing general circulation model (GCM) [18]. In the context of short-term heavy rainfall prediction, support vector machine (SVM), k-nearest neighbor's algorithm (k-NN), and variant k-NN (k-VNN) have been evaluated. Notably, SVM with a polynomial kernel demonstrates favorable equitable treatment scores, while k-VNN showcases improvement over conventional k-NN, albeit surpassed by the SVM model [19]. Furthermore, the entropy and gamma test (GT) approach for long-term rainfall prediction outperforms the entropy method in identifying optimal input combinations for models [5]. The mutual information (MI) technique is a significant factor in selecting explanatory variables for ANN modeling, resulting in augmented model performance through simplified architecture [20]. Lastly, a study focusing on monthly rainfall prediction employs Pearson's correlation to identify IVS methods, showcasing the significance of sea surface temperature (SST) in predicting rainfall, followed by other climatic variables. This method displays the potential for optimized predictive performance when utilizing SST as input [6]. All examined studies underscore an evident necessity for developing more robust IVS methodologies that possess attributes of computational efficiency, independence from prior information or preconceived conceptions regarding the system, and the capability to elucidate non-linear and interconnected relationships among potential input variables. This study addresses the complex challenge of input selection in three distinct facets within the context of identification: (a) the most efficacious predictors, (b) the optimal combination of input variables for monthly rainfall forecasting, and (c) understanding the impact of large-scale climatic variables on rainfall forecasting. A comparative analysis is conducted between the performance of the support vector regression-recursive feature elimination (SVR-RFE), Gradient boosting (GB), and proposed bootstrapped long-short-term memory recurrent neural network (BTSP-LSTM-RNN). These models are evaluated by statistical metrics such as coefficient of determination (R^2), mean absolute error (MAE), root mean squared error (RMSE), Pearson's correlation coefficient (r), and mean absolute percentage error (MAPE).

The scope of this investigation encompasses two different basins within Thailand. The main objective sustaining this research, particularly concerning models employed in rainfall forecasting, pertains to reducing input data requirements, mitigating predictive uncertainty, and optimizing computational resources necessary for these models. By achieving these goals, an advancement in the accuracy of rainfall forecasting is anticipated. The organizational structure of the manuscript is as follows: the 'Materials and Methods' section expounds upon the employed river basins, dataset, and IVS methodologies, encompassing SVR-RFE, GB, and BTSP-LSTM-RNN and prediction models, including LSTM-RNN, MLP-ANN, and BTSP-RF. The outcomes garnered from this empirical investigation are delineated in the ensuing section, culminating in a concise summary and conclusion. The overall methodology employed for the IVS is described in Fig. 1.

2. Materials and methods

2.1. Study area

Thailand is divided into five climatic regions. The northern region, characterized by extensive mountain ranges, is the source of four major river basins: the Ping, Wang, Yom, and Nan rivers. These waterways converge to form the Chao Phraya River Basin, an economic epicenter of

Table 1

Overview of some input variable selection studies for rainfall forecasting in different regions of the World.

Method (s)	Forecasting Type	Dataset	Data Frequency	Source of Data	Study Area/ Country	Evaluation Criteria	Outcomes	Ref.
GA and SOM with the application of winGamma software	Long-term rainfall forecasting	Observed rainfall and temperature data and large-scale climatic signals	30 years (1979–2014)	NASA National Weather Climate Center	Different stations in Iran	NSE	The GA approach exhibits heightened dependability in comparison to the alternative paired methods. Its applicability extends to prospective employment as a dependable means of IVS. Furthermore, amid the array of diverse climatic signals considered, namely the PDO, TNI, and SST-Nino 3, these indices manifest as the most recurrent and pivotal for achieving optimally precise prognostications for each respective observational station.	[17]
Circle Inspired Optimization Algorithm-Deep LSTM	Rainfall Forecasting	Past Rainfall Time Series Dataset	Not Available	Weather stations in Australia	Australia	RAE, MSE, and RMSE	The empirical findings demonstrate that the formulated Inspired Optimization Algorithm-Deep LSTM paradigm yields heightened performance levels, as evidenced by its minimal RAE value of 0.023, MSE registering at 0.151, and RMSE measuring 0.389.	[1]
Monthly Rainfall Forecast Model Deep Neural Networks (MRDNN)	Monthly Rainfall Forecast	LAV and Meteorological Dataset (Monthly Rainfall).	Monthly rainfall data from 1991 to 2016	Earth System Research Laboratory (ESRL) and Thai Meteorological Department (TMD)	Khleng Yai river basin, eastern Thailand	Stochastic efficiency	The proposed DNN monthly rainfall forecast model can provide an acceptable accuracy of forecast about 70% up to for 3-months leading time of forecast.	[7]
ANN model with climatology and Predictive Ocean Atmosphere Model for Australia (POAMA)	Monthly Rainfall Forecast	Observed historical rainfall, pressure, maximum, minimum temperature, and climate indices, i.e., SOI, DMI, Nino 3.4, SAM, IPO, and DMI, SST1, and SST2	There are different starting years for each data, but the ending year is the same for all dataset	Nino 3.4 and DMI from the Royal Netherlands Meteorological Institute (KNMI) Climate Explorer, IPO data from Chris Folland, Met Office, Hadley Centre, UK, SAM data from http://jisao.washington.edu/data/aao/slp , SST1 and SST2 from http://www.bom.gov.au/climate/ahead/sst_data_table.html .	Queensland, Australia	RMSE, MAE, and r	The utilization of ANN for predictive modeling across three geographically distinct regions within Queensland has yielded demonstrably enhanced forecast outcomes. These advancements are particularly discernible through the attainment of reduced RMSE, MAE, and elevated (r) when contrasted with the prognostications generated by the POAMA. Notably, POAMA is the prevailing GCM employed for seasonal rainfall forecasts.	[18]
SVM, k-NN, and k-VNN	Short-Term Heavy Rainfall Prediction	Real weather data was collected from 408 automatic weather stations.	2007–2010	Automatic weather stations of Seol	Seoul, South Korea	ETS, FBI, PC, POD and PAG	The SVM employing a polynomial kernel exhibited the most elevated equitable treatment score on average. Furthermore, the k-VNN algorithm	[19]

(continued on next page)

Table 1 (continued)

Method (s)	Forecasting Type	Dataset	Data Frequency	Source of Data	Study Area/ Country	Evaluation Criteria	Outcomes	Ref.
Entropy and Gamma Test (GT)	Long-term rainfall prediction	Monthly rainfall data, monthly SLP, difference in sea level pressure (DSLP), and sea surface temperature (SST)	1970–2010	Iran's Meteorological Organization and National Center for Atmospheric Research.	Aharchay river basin, northwestern part of Iran	RMSE, r, and NSE	demonstrated superior performance to the conventional k-NN approach. However, it is noteworthy that despite this improvement, the performance of k-VNN was surpassed by the SVM model utilizing a polynomial kernel. The GT performed better in identifying a model's optimal combination of inputs than the entropy method.	[5]
Mutual information (MI) technique	Rainfall Forecasting	Historical rainfall, cloud images, atmospheric pressure, dry bulb temperature, relative humidity, and wind speed dataset.	1998–2006	Indian Meteorological Department	Mumbai, India	Efficiency index (EI), RMSE and NRMSE	This investigation underscores the superiority of the MI technique in selecting explanatory variables for ANN modeling. This superiority is not solely attributable to the augmented model performance across diverse indicators but also to achieving this elevated performance by utilizing a simplified ANN architecture.	[20]
Pearson's correlation was used between climate variables and rainfall to identify the IVS method for forecasting models.	Monthly Rainfall Prediction	Monthly observed rainfall, dry bulb temperature, relative humidity, and wind speed dataset. Large-scale climatic variables (Southern Oscillation Index (SOI), SST, and Indian Ocean Dipole Mode Index (DMI) and NINO1 + 2, NINO3, NINO3.4, and NINO4.	TMD, National Oceanic and Atmospheric Administration (NOAA)		Thale Sap Songkhla River Basin, Thailand	Overall index (OI), RMSE, r, and MAE.	The correlation analysis exhibited that large-scale climatic variables significantly impact the monthly rainfall prediction within Thailand's Thale Sap Songkhla basin. Sequentially, SOI and DMI followed in terms of influence. Furthermore, it is notable that the models constructed utilizing SST as input variables yielded optimal predictive performance across most examined models.	[6]

Thailand, located within the central plains and serving as its largest agricultural zone. The southern region, marked by prolonged mountain ranges and steep gradients, features the Tapi River and the Andaman Sea, contributing to its allure as a regional tourist destination. In the northeastern region, primarily a plateau, the landscape accommodates three prominent rivers—the Mae Klong, Chi, and Mun Rivers—supporting vital agricultural and cultivation activities. Renowned for its high-quality jasmine rice, this area also confronts economic challenges, rendering it one of Thailand's less affluent regions. Anchored in distinct geographical domains, Thailand is divided into 22 individual river basins presented in Fig. 2 [21,22].

2.2. Dataset analysis

This investigation strategically selected two basins from the northern region. These selections were based on the criteria of having the lowest and highest mean annual rainfall over the past three decades, from 1993 to 2022. Table 2 presents a detailed characteristic of both selected River basins. The Wang River basin, in the northern region, exhibits the lowest

mean annual rainfall of 3898.47 mm and a standard deviation (SD) of 934.62 mm. This level of rainfall variance reflects the basin's hydro-climatic conditions, which have been pivotal in shaping its distinctive rainfall patterns. The basin's rainfall ranges from a minimum of 2735.55 mm to a maximum of 6453.26 mm, as observed across its three hydro-meteorological monitoring stations. The Nan River basin with mean annual rainfall of 9357.04 mm, exhibiting a marked difference from the Wang River basin. With an SD of 2105.61 mm, it experiences considerable annual rainfall variability, likely due to regional geographical and climatic factors. This basin's recorded rainfall ranges from a minimum of 6718.54 mm to a maximum of 14,807.91 mm, highlighting its susceptibility to drought and heavy rainfall events. These rainfall characteristics significantly affect the basin's hydro-climatic conditions and water resource management.

This study has used meteorological datasets comprising monthly rainfall (RA), maximum and minimum temperatures (Tmax and Tmin), relative humidity (RH), wind direction (WD), and wind speed (WS). These datasets were sourced from the Thai Meteorological Department (TMD) station within the Wang and Nan River basins. Fig. 3 shows the

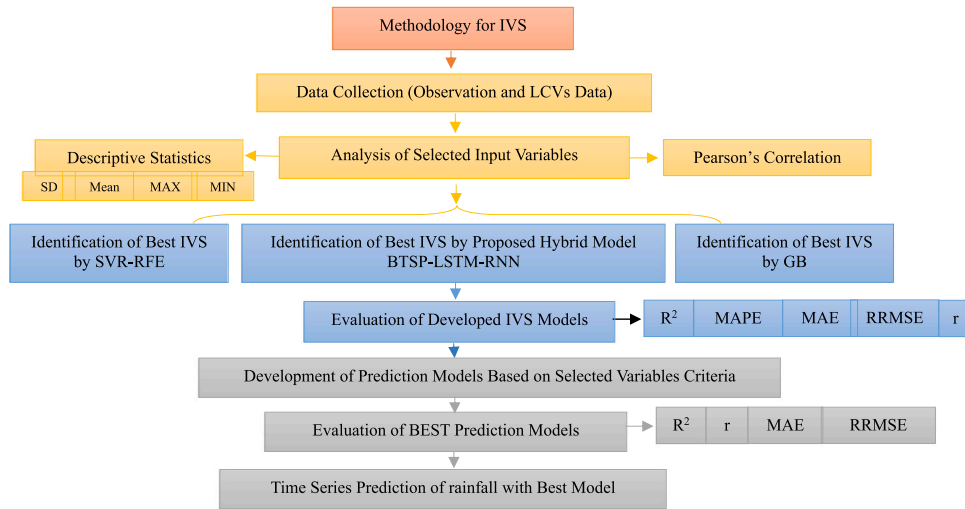


Fig. 1. Overall methodology for input variable selection, Input variable combinations, and rainfall prediction.

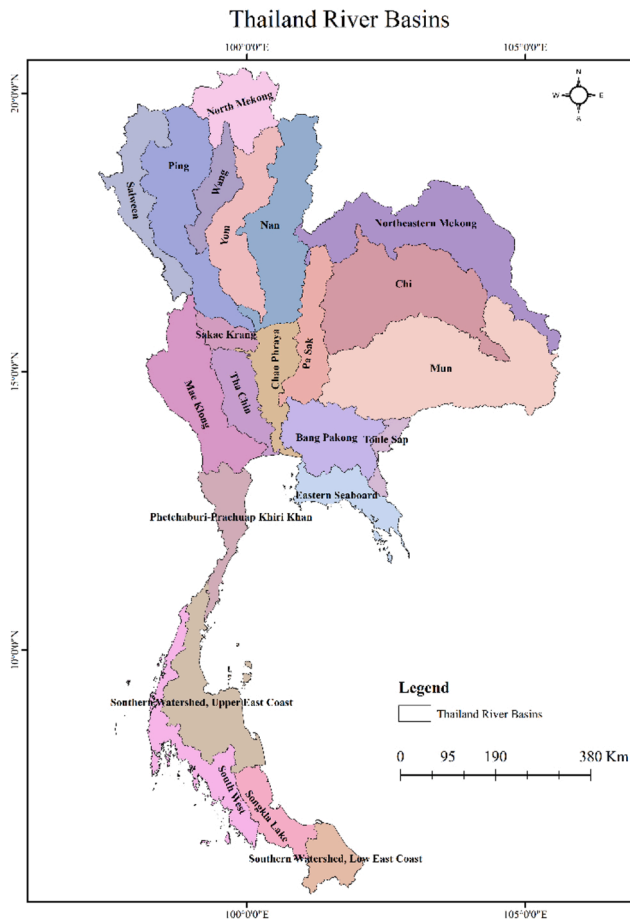


Fig. 2. River Basins in Thailand.

distribution of Thai Meteorological Department (TMD) stations over selected River basins in the Northern regions of Thailand. It is worth noting that certain variables, specifically solar radiation (SR) and surface pressure (PS), were not accessible due to either inadequate data quality or unavailability. Consequently, we obtained supplementary data for these variables from the National Aeronautics and Space Administration (NASA) [23]. Furthermore, this study incorporated three extensive monthly oceanographic index datasets spanning 1993–2022.

Specifically, these datasets encompassed the Southern Oscillation Index (SOI), Sea Surface Temperature (SST), and Indian Ocean Dipole Mode Index (DMI). The SOI quantifies variations in atmospheric pressure above the sea surface between the locations of Darwin and Tahiti within the Pacific Ocean. SST measurements in the central Pacific Ocean were obtained from distinct regions known as NINO1 + 2 (0–10 S, 90–80 W), NINO3 (5S–5 N, 150–90 W), NINO3.4 (5S–5 N, 170–120 W), and NINO4 (5S–5 N, 160–150 W), as provided by the National Oceanic and Atmospheric Administration (NOAA) website. Prior research has demonstrated that these particular indices exhibit a stronger correlation when compared to other indices such as DMI, SST, and SOI[24].

The meteorological dataset contains missing values estimated by the K-nearest neighbor method [25]. Figs. 4 and 5 shows the descriptive statistics of all input variables at the Wang and Nan River basins. The statistical analysis of the dataset reveals significant variations in several key parameters. At the Wang River basin, the mean RA of 324.87 mm/month indicates the average amount of rainfall received, which is crucial for assessing water availability. The SD of 290.75 mm/month suggests a considerable variability, highlighting the need for robust water management strategies. The RH of 73.33% and SD of 7.94% imply moderately humid conditions with some seasonal fluctuations. WD and WS, with mean values of 177.75 degrees and 9.87 m/s, respectively, reflect the prevailing wind characteristics that influence evaporation rates and local microclimates. Temperature data, including mean Tmax at 33.41 °C and mean Tmin at 21.33 °C, are essential for assessing temperature-related impacts on water resources, such as evapotranspiration rates. The SR mean of 98.01 W/m² indicates that incoming energy affects local climate and hydrological processes. Lastly, the oceanic indices, including the SOI and various Nino indices, indicate large-scale climate patterns influencing regional climate variability. At the Nan River basin, the mean monthly RA of 779.75 mm reveals the average rainfall received, indicative of the region's hydrological regime. The relatively high SD of 676.29 mm underscores substantial variability in rainfall, reflecting the basin's susceptibility to wet and dry periods. Mean RH at 76.59% and low SD of 5.68% denote consistently humid conditions essential for ecosystem stability and groundwater recharge. Wind characteristics, represented by a mean WD of 179.57 degrees and SD of 18.14 degrees, alongside a mean WS of 6.64 m/s and its higher variability, influence evaporation and microclimates. Temperature data, with mean Tmax at 32.94 °C and mean Tmin at 21.35 °C, underscore the region's temperature dynamics affecting evapotranspiration rates and seasonal variations in water availability. SR data indicate an average energy input of 97.24 W/m², influencing temperature patterns and hydrological processes. In contrast, the Nan basin exhibits substantially

Table 2
Overview of selected River basins, annual rainfall statistics, and basin’s unique characteristics.

River Basins	Region	No. of TMD Stations	Mean (mm)	SD (mm)	Min (mm)	Max (mm)	Basin’s unique characteristics (Climatic, Geographical, and Hydrological)
Wang River Basin	North	3	3898.47	934.62	2735.55	6453.26	Discernible climatic, geographical, and hydrological characteristics characterize the Wang River Basin. Influenced by a tropical monsoonal climate, the basin undergoes seasonal oscillations between wet and dry periods, profoundly impacting its hydrological dynamics. Encompassing a defined area, predominantly within northern Thailand, the basin’s geographical attributes influence its local significance and resource utilization patterns. The Wang River, a prominent tributary of the Chao Phraya River, assumes a crucial role in regional water provisioning, agricultural sustenance, and overall hydrological connectivity, warranting prudent water resource management strategies.
Nan River Basin	North	6	9357.04	2105.61	6718.54	14,807.91	The Nan River Basin exhibits discernible climatic, geographical, and hydrological attributes integral to its ecosystem dynamics. Subject to a tropical monsoonal climate, the basin’s hydrological patterns are distinctly characterized by alternating wet and dry phases, influencing water resource availability. Primarily situated in northern Thailand, the basin’s geographical context underscores its regional significance and environmental interactions. The Nan River, a principal tributary of the Chao Phraya River,

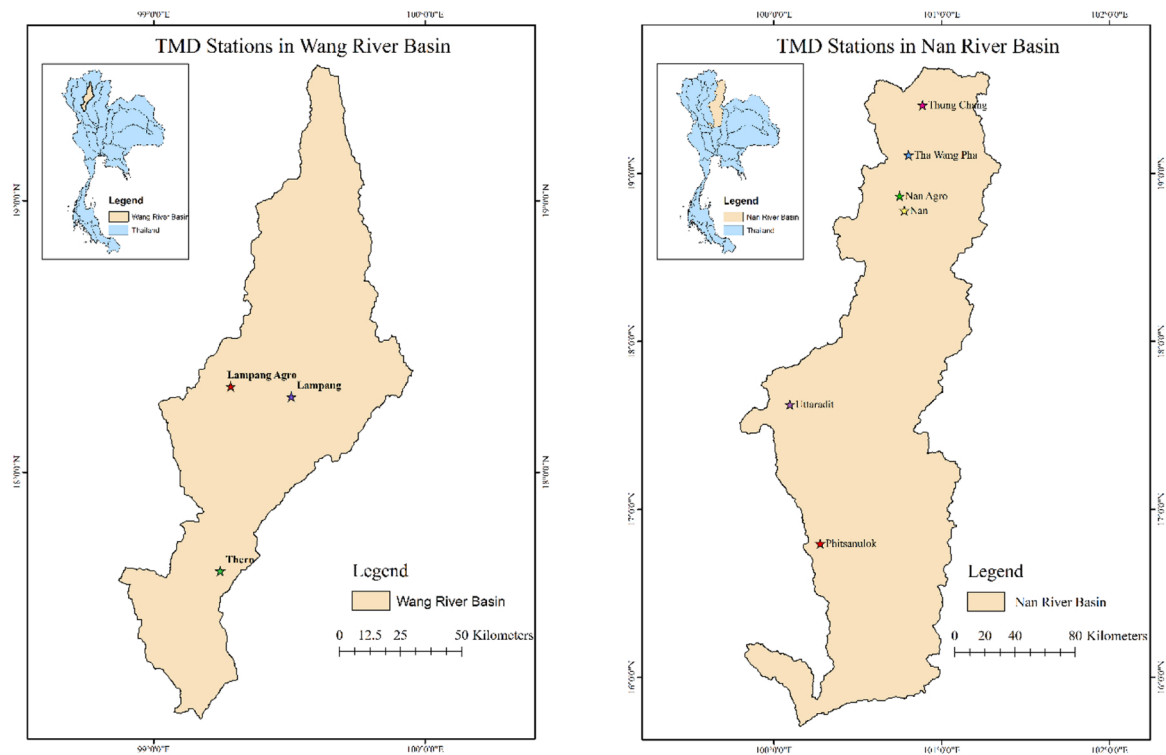


Fig. 3. Distribution of TMD stations over selected River basins in the Northern Region.

higher rainfall levels, with a mean of 9357.04 mm and a standard deviation of 2105.61 mm. This basin encompasses six monitoring stations, and its annual rainfall varies from a minimum of 6718.54 mm to a maximum of 14,807.91 mm. These statistics offer valuable insights into the rainfall characteristics of these river basins, which can be instrumental in hydrological and environmental research and management efforts within their respective regions.

Pearson’s correlation coefficient (r) is a statistical measure to assess the linear relationship between two continuous variables. Fig. 6 presents a correlation between RA and other meteorological and LCVs at the Wang River basin. The RA and RH are approximately 0.538, indicating a moderate positive linear association. When RH increases, there is a tendency for rainfall to increase. This relationship suggests that higher humidity levels contribute to greater chances of rainfall. The correlation between RA and WS is approximately 0.332, indicating a moderate positive linear association. Higher wind speeds are moderately

correlated with increased rainfall, suggesting that stronger winds can contribute to greater rainfall. RA exhibits a positive correlation with Tmax and Tmin, with correlation coefficients of approximately 0.079 and 0.684, respectively. These correlations suggest that higher temperatures, both maximum and minimum, are associated with increased rainfall. This relationship aligns with the expectation that warmer temperatures lead to more evaporation and atmospheric moisture, which can result in greater rainfall. The correlation coefficient between RA and WD is approximately 0.223, suggesting a weak positive linear relationship. It implies that wind direction has a minor influence on rainfall patterns in this dataset. The correlations between RA and various climate indices (SOI, Nino 1.2, Nino 3, Nino 3.4, Nino 4) and other meteorological variables (SR, PS, and DMI) vary in strength and direction. These correlations provide insights into how these climate and meteorological factors are related to rainfall in the specific dataset.

Fig. 7 shows the correlation coefficients between RA and other

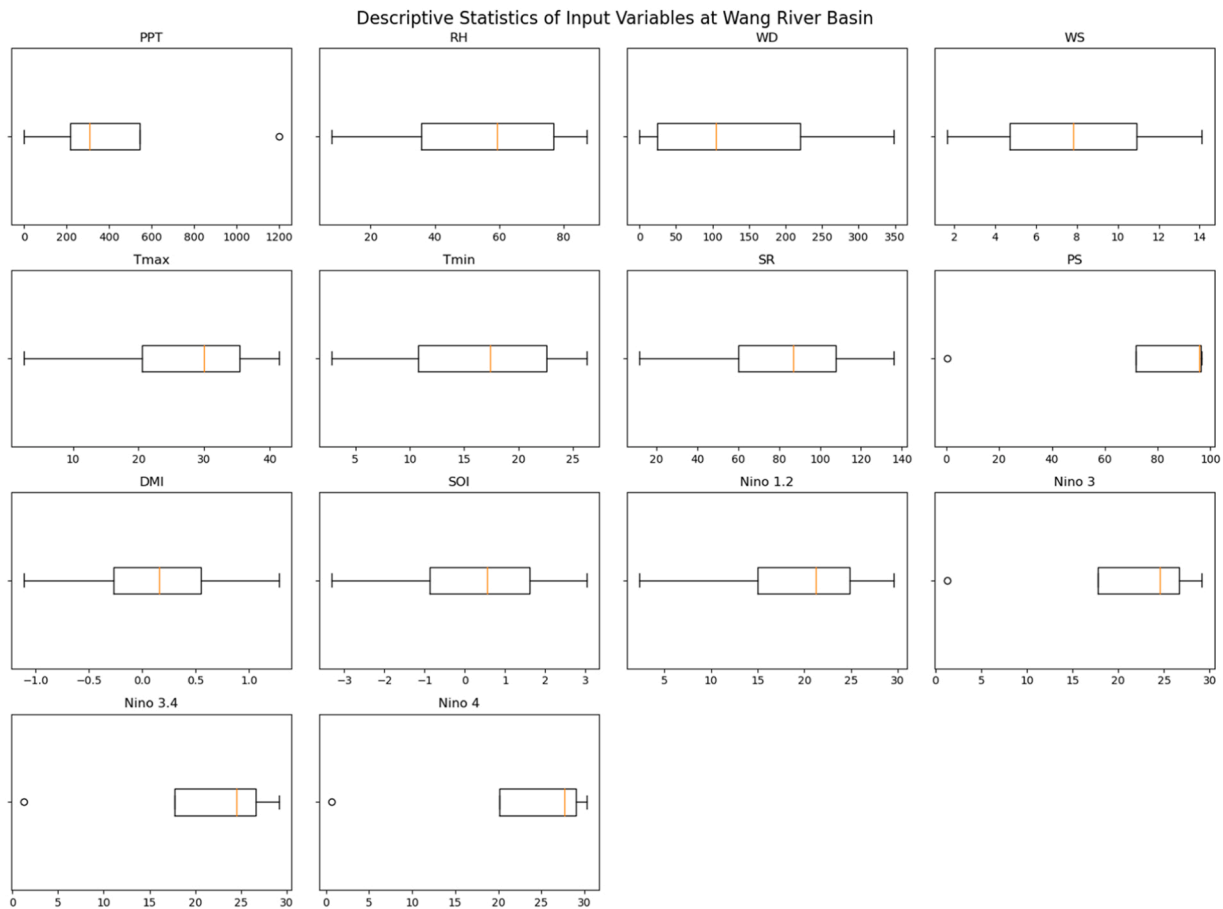


Fig. 4. Descriptive statistics of input variables at Wang River basin.

meteorological and LCVs at the Nan River basin. The correlation between RA and RH is notably positive, with an (r) of 0.671. It indicates a moderate positive linear association between RA and RH. When RH increases, rainfall tends to increase, which is an expected relationship as higher humidity levels often lead to greater chances of rainfall. In contrast, the correlation between RA and WD is weak, with an (r) of 0.209. It suggests only a slight positive linear relationship between RA and WD. The direction of the wind may not strongly influence rainfall patterns in this context. Similarly, the correlation between RA and WS is also weak, with an r-value of approximately 0.207. It implies a minor positive linear relationship between RA and WS, indicating that wind speed does not substantially impact rainfall in this dataset. Temperature variables, Tmax and Tmin, show moderate positive and strong positive correlations with RA, with r-values of approximately 0.207 and 0.733, respectively. It suggests that as temperatures increase, rainfall tends to increase as well. Higher temperatures can increase evaporation and atmospheric moisture, resulting in more rainfall when other factors align. The correlation coefficients between RA and other climate indices such as the SOI, Nino indices (Nino 1.2, Nino 3, Nino 3.4, Nino 4), and other meteorological variables (SR, PS, and DMI) indicate both positive and negative relationships over Nan River basin.

2.3. Input variable selection (IVS) techniques

In dimensionality reduction techniques, two principal categories emerge: feature selection and dimensionality reduction. While both approaches share the overarching goal of reducing the dimensionality of data, they diverge in their fundamental methodologies. Feature selection, as its name implies, focuses on identifying pertinent features within the dataset, opting to retain variables in their original format. In

contrast, dimensionality reduction involves a transformative process in which variables are merged or otherwise modified [26]. Given the continuous nature of the data under examination, methodologies adept at handling continuous variables are deployed as part of this investigative endeavor. Within this domain of feature selection, ensemble techniques (GB, SVR-RFE, and BSTP-LSTM-RNN) are employed in the present study.

2.3.1. Gradient boosting (GB)

The Gradient Boosting Regressor (GBR) represents an important ensemble model characterized by its iterative nature, wherein a series of sequentially arranged tree models are employed. This distinctive design allows each subsequent model to learn from the errors made by its predecessor, thereby enhancing the overall predictive capacity [27]. Within this ML framework, predictions are generated through "boosting," which involves aggregating relatively weak predictive models, frequently in the form of decision trees [28]. The cumulative effect of this ensemble approach results in creating a more resilient and robust predictive model [29]. Mathematically, M' iterations of tree models can be represented as follows:

$$f_M(x_i) = \sum_m \gamma_m h_m(x_i) \quad (1)$$

' h_m ' represents a weak learner, characterized by its relatively modest individual performance, while ' γ_m ' serves as a scaling factor, augmenting the influence of a tree within the model [29]. The GB strategically employs a gradient descent loss function to systematically mitigate errors by iteratively refining initial estimations with updated ones. Consequently, a final predictive model emerges, synthesizing the collective contributions of all preliminary estimations while assigning

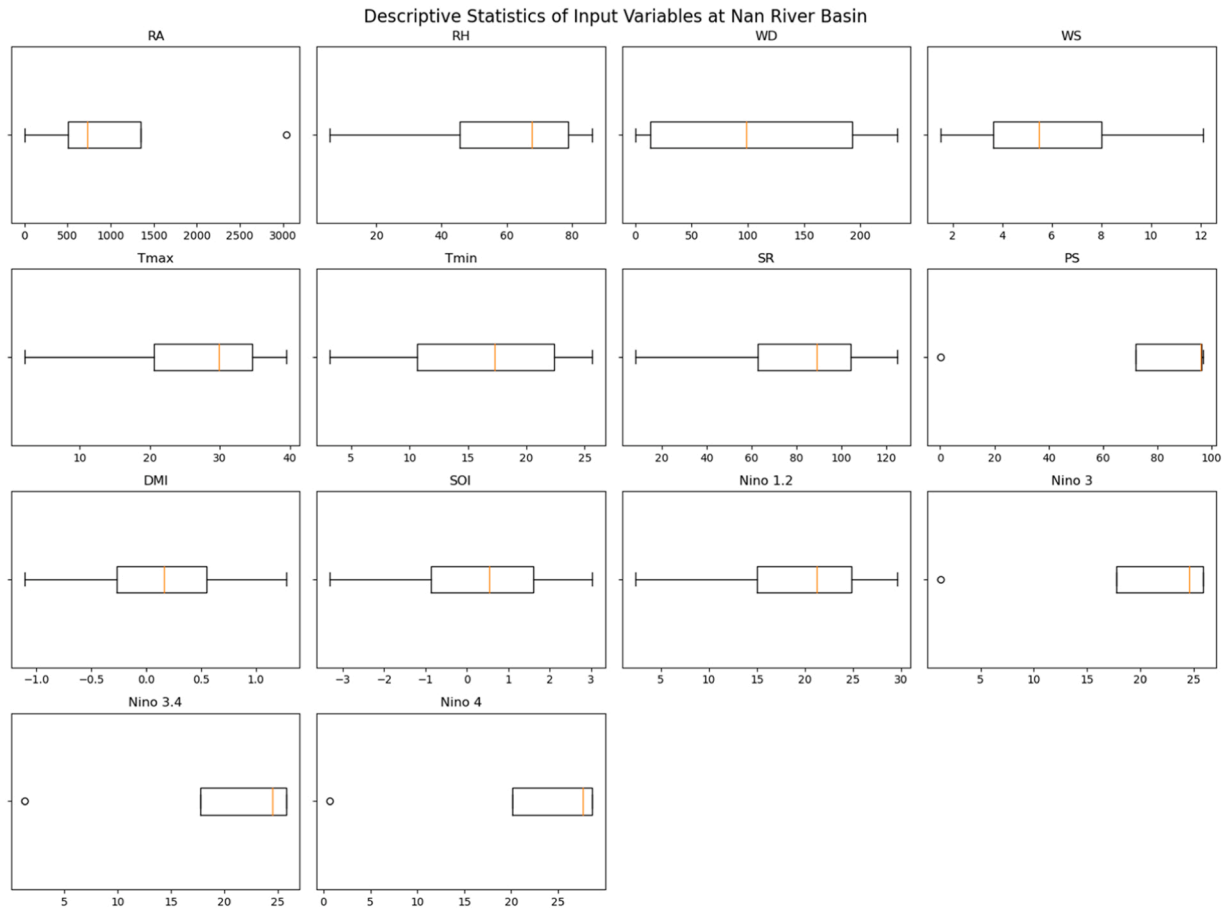


Fig. 5. Descriptive Statistics of input variables at Nan River basin.

appropriate weights to each. The GBR model utilized in this research is instantiated through the 'GradientBoostingRegressor()' method, as made available in the Scikit-learn framework [30]. In this study, GBR operates by sequentially training decision tree regressors, with each subsequent tree designed to rectify the errors made by the preceding ones. The model is initialized with default hyperparameters, including the number of trees (estimators = 100), the learning rate (learning rate = 0.01), and the maximum depth allowed for individual decision trees (Max_depth = none). Hyperparameter tuning is executed via a grid search to optimize the model's performance. It entails exploring various combinations of hyperparameters to identify the set that yields the most favorable predictive outcomes. The hyperparameter space under investigation encompasses three values for 'estimators' (100, 400, 1000), three values for 'learning rate' (0.01, 0.1, 0.5), and three values for 'max_depth' (3, 4, 5), resulting in a total of 27 distinct hyperparameter configurations. During the grid search, a 5-fold cross-validation strategy is employed. This method partitions the dataset into five subsets, with each subset serving as the test data once while the model is trained on the remaining portions. Following the grid search, the optimal hyperparameters are discerned, revealing that the model performs optimally with a learning rate of 0.01, a maximum depth of 3 for decision trees, and a substantial ensemble of 1000 trees.

2.3.2. Support vector regression with recursive feature elimination (SVR-RFE)

In 2002, Guyon introduced SVM-RFE as a technique for feature selection. SVM-RFE is an iterative feature selection method that relies on SVM weights for feature ranking [31]. This method is categorized as an embedded feature selection technique and has found widespread application, particularly in gene selection. [32]. The core concept

behind SVM-RFE is to iteratively remove features with the lowest weight values [31]. In this study, we employed SVR for the meteorological and LCVs continuous dataset. It is an ML model used for regression tasks, where the goal is to predict a continuous numerical output [33]. SVR-RFE aims to select a subset of the most relevant features from a dataset to improve the accuracy of an SVR model. Like SVM-RFE, SVR-RFE also ranks features based on their importance and iteratively eliminates less important ones [34]. The general procedure used for RFE is as follows: (1) Train a classifier to determine the weight vector (w), (2) compute ranking criteria for all features, and (3) eliminate features with the smallest ranking criterion value. The elimination of features in each iteration is carried out through a backward feature elimination process. The ranking score is determined by the components of the SVR's weight vector (w) as:

$$w = \sum_k a_k y_k x_k \quad (2)$$

Algorithm used for SVR-RFE:

Input: training data ($\{x_i, y_i\}_{i=1}^N$).

Output: ranked feature list R.

Initialize:

$S = \{1, 2, \dots, D\}$.

$R = \emptyset$.

While S is not empty, do:

1. Restrict the features of to X_j the remaining S.

2. Train SVR to get weight vectors.

3. Calculate the ranking criteria $c_k = w_k^2$, $k = 1, 2, \dots, |S|$.

4. Look for features with the smallest value of c_k called feature p.

5. Add feature p into R ($R = \{p\} \cup R$).

6. Remove feature p from R ($S = S/p$).

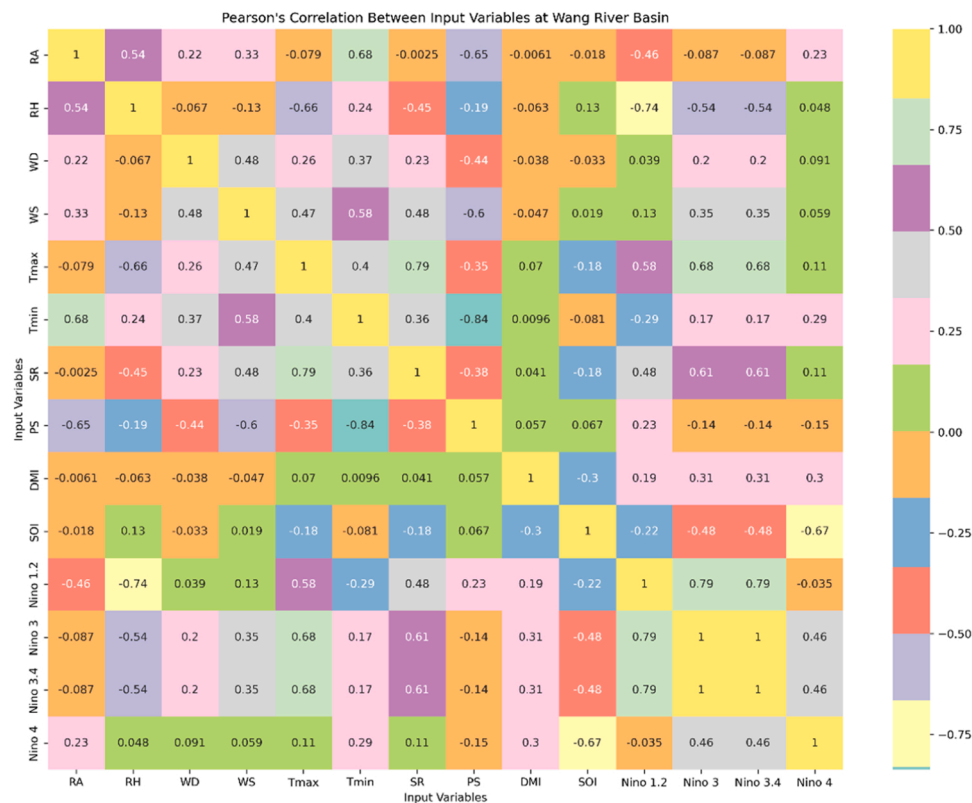


Fig. 6. Pearson's correlation between input variables at the Wang River basin.

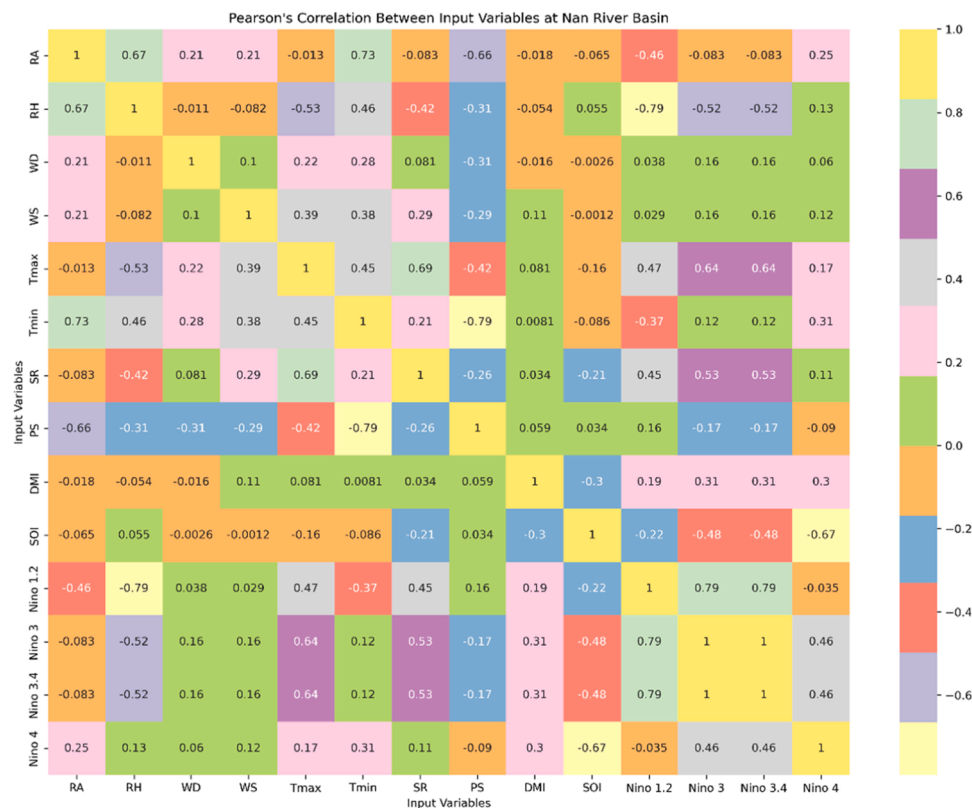


Fig. 7. Pearson's correlation between input variables at Nan River basin.

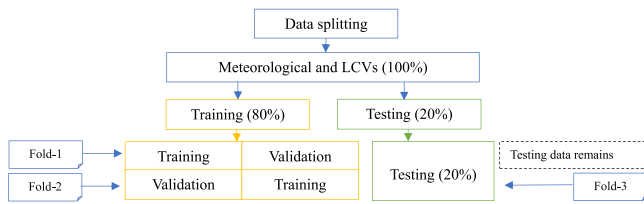


Fig. 8. Input variables dataset splitting for training and testing of SVR-RFE.

Table 3
Hyperparameters tuning for the SVR model.

Hyperparameter	Values
C	0.1, 1, 10, 100, <u>1000</u>
Kernel	Linear, polynomial, <u>rbf</u> , and sigmoid
epsilon	0.001, 0.01, 0.1, 0.2, 0.5, <u>1.0</u>
Cross-validation	2, 3, 5-fold

In this study, 80% of the dataset was used to train and validate SVR-RFE, and 20% was used for testing, as shown in Fig. 8.

we employed a hyperparameter grid search to identify the optimal hyperparameters for the SVM-RFE model [35]. As summarized in Table 3, the hyperparameter grid was pivotal in this process. Notably, the "C" parameter was a key focus, governing the balance between correctly classified instances and the hyperplane's capacity to separate them. It's worth highlighting that the "rbf" kernel was chosen due to its unique ability to produce feature importance values, which is crucial for the RFE algorithm's feature ranking. Additionally, we considered the "step" hyperparameter, which determines the number of features to eliminate in each iteration. A 3-fold cross-validation strategy was selected after 2,3 and 5-fold cross-validation was employed earlier during the hyperparameters tuning. This method partitions the dataset into three subsets, each serving as the test data once while the model is trained and validated on the remaining portions. Following the grid search, the optimal hyperparameters are discerned, revealing that the model performs optimally with a "C" of 1000, and a 3-fold cross-validation strategy ensures more generalized results. Ultimately, the SVM-RFE algorithm yielded 14 features with the most relative importance towards the target (RA). The final hyperparameters of the SVR model are underlined in Table 3.

2.3.3. Bootstrapped long short-term memory recurrent neural network (BTSP-LSTM-RNN)

This study developed an innovative approach to enhance rainfall forecasting by identifying and selecting relevant input features using a combination of LSTM-RNN and bootstrapping. The recurrent neural network (RNN) originates in the Hopfield network, introduced by Hopfield in 1982. It represents a distinct neural network structure conceived under the premise that human cognition relies on experience and memory [36]. With the evolution of deep learning and the advent of Big Data, researchers have recognized the RNN's remarkable data

mining capabilities, leading to widespread adoption. Unlike traditional feed-forward neural networks (FFNNs), RNNs feature connections between neurons across layers, a crucial aspect for tasks involving time series prediction. This interconnection enables the current output to be influenced by previous outputs. Each neuron within each layer is not isolated but possesses a directional cyclic relationship. Consequently, RNNs retain and utilize past information in computing present outputs [37]. Fig. 9 shows the general process of feed-forward propagation in an RNN through folded and unfolded visual representations. LSTM was proposed by Hochreiter and Schmidhuber [38] as a type of network developed from RNNs. However, RNN can only view historical data for a short time. Therefore, it is not powerful enough to learn patterns from long-term dependencies. This will cause issues with backpropagation because it goes back to many steps and nodes. As a result, the vanishing gradient problem occurs. Therefore, the LSTM technique was explicitly created for learning long-term dependencies by designing 24 functional parts of the memory cell state to solve and overcome the weaknesses of traditional RNNs [6] (Fig. 10).

Bootstrapping, introduced by Efron (1979), has become a fundamental technique in statistical inference [39]. It offers a robust and versatile approach to estimating the distribution of a statistic without relying on strong distributional assumptions [40]. The core idea is to repeatedly resample, with replacement, from the observed data to create many bootstrap samples. By analyzing these samples, researchers can make inferences about the population from which the data is drawn [41]. Bootstrapping, a resampling technique that involves drawing random samples with replacement from the dataset, can be integrated with LSTM RNNs to enhance input variable selection[42]. Bootstrapping generates multiple subsets of the data, each representing a different variation of the original dataset. LSTM-RNN models can be trained on these bootstrapped samples, enabling the assessment of input feature importance across multiple models [43]. The details about LSTM-RNN architecture and mathematical terms are explained in Section (2.4.3).

To identify the importance of input variables, BTSP-LSTM-RNN models can be analyzed. Techniques such as analyzing the weights of connections between input features and LSTM cells or employing permutation importance can help quantify the contribution of individual variables to the model's forecasting performance. This methodology involves several key steps. Firstly, preprocess the data by scaling the input features and employing bootstrapping to generate multiple samples. Bootstrapping, a resampling technique involving randomly drawing samples with replacement, allows the model to create diverse datasets for evaluating feature importance. Secondly, design an LSTM-RNN architecture. Table 4 and Fig. 10 presents the final hyperparameters for the BTSP-LSTM-RNN model. This architecture comprises two LSTM layers, each with 50 neurons, followed by a dense output layer. While this represents a basic LSTM model structure, hyperparameter tuning and architectural adjustments may be necessary for optimal results. Data transformation is crucial for LSTM networks, and we achieve this by creating lagged data sequences. This format ensures that the LSTM can effectively capture temporal dependencies in the rainfall data. Subsequently, we proceed to the core of our approach,

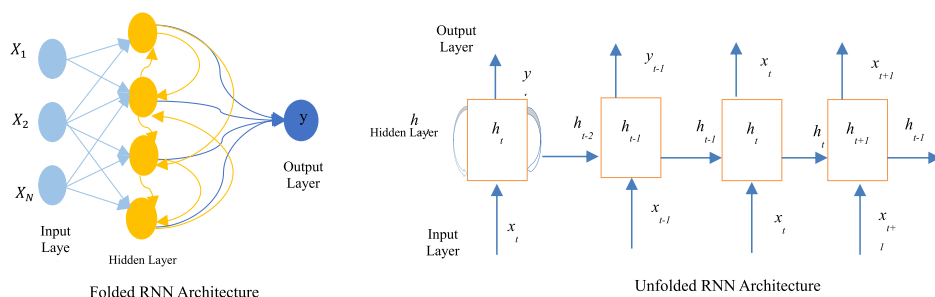


Fig. 9. General folded and unfolded structure of the recurrent neural network.

Model: "sequential_9"

Layer (type)	Output Shape	Param #
lstm_18 (LSTM)	(None, 1, 50)	12800
lstm_19 (LSTM)	(None, 50)	20200
dense_9 (Dense)	(None, 1)	51

=====
Total params: 33051 (129.11 KB)
Trainable params: 33051 (129.11 KB)
Non-trainable params: 0 (0.00 Byte)

Fig. 10. LSTM-RNN model architecture consists of two LSTM layers and 1 Dense Layer.

Table 4
Final Hyperparameters of BTSP-LSTM-RNN model.

Hyperparameters	Values
No. LSTM layers,	2
No. of neurons	50
Dense layer	1
Optimizer	Adam
No. of Bootstrap iterations	10
Loss	MSE

which involves bootstrapping to train multiple LSTM models on the generated datasets. We iterate this process a specified number of times, such as 10, to obtain a collection of LSTM models, each trained on a distinct sample.

2.4. Monthly rainfall prediction models

2.4.1. Bootstrap aggregation on Random Forest (Ensemble Method)

Random Forest (RF) emerges as a highly efficient ML model for predictive analytics, celebrated for its assembly of individual decision trees (DTs). RF's effectiveness stems from the fusion of two fundamental elements: (1) randomness and (2) ensemble learning, concepts elucidated by Breiman (2001, 2003) [44,45]. Ensemble ML entails the fusion of diverse models, each contributing distinct insights, to collectively provide a holistic perspective on the target variable [46]. In the context of this analysis, our study evaluated an ensemble ML approach by combining extreme"-gradient-boosting and RF [30]. The strength of RF lies in its strategic utilization of randomness during the construction of individual decision trees within the ensemble. This randomness manifests in two critical aspects: selecting data samples and choosing input features. The RF process commences by randomly sampling the dataset. More precisely, from a dataset of size "N," a subset of size "N'" containing "m" variables is randomly selected. Subsequently, a single decision tree is formed. Notably, the tree-building process adheres to the classification and regression tree (CART) algorithm [47]. This investigation employed meteorological and LVCs data, encompassing two input variable combinations. To construct each DT, we meticulously performed hyperparameter tuning by grid research to determine the optimal parameters shown in Table 5.

The grid search gives the best parameters as min samples leaf= 4, n estimators (K) = 300, and random state= 42 for given IVCs. This procedure is reiterated K times, resulting in the creation of an ensemble comprising K trees. These K trees collectively contribute to the final predictions for input data, exemplifying the essence of ensemble learning. Each decision tree within the ensemble plays a pivotal role in the ultimate prediction: In classification tasks, the final predicted class is determined through the majority rule, with the class accumulating the

Table 5
Hyperparameters Tuning for prediction models.

Hyperparameters Tunning for Random Forest	
estimators	[100, 200, 300]
max_depth	[None, 10, 20, 30]
min_samples_split	[2,5,10]
min_samples_leaf	[1,2,4]
Hyperparameters Tunning for MLP-ANN	
hidden_layer_sizes	[(50,),(100,),(50,50),(100,50)]
activation	['relu', 'tanh']
alpha	[0.0001, 0.001, 0.05]
learning_rate	['constant', 'adaptive']
max_iterations	1000
Hyperparameters Tunning for LSTM-RNN	
No. of hidden nodes	[10,20,30,40,50]
learning_rate	[0.1, 0.01, 0.001, 0.05, 0.005]
Dense layers	[5,10,20,30,40]
Activation function at each layer	['tanh', 'relu', 'sigmoid']
Optimizer	Adam
Verbose	[1,2]
Batch Size	[12,32,64]
Epoch	[100, 300, 500, 800, 1000]

highest number of votes designated as the ultimate prediction. In regression tasks, the final predicted value is calculated by averaging the outcomes from all individual DTs. The implementation of RF in the current study was facilitated using Python 3.11, with the typical choice of K set to 300. Generally, larger values of K tend to enhance model performance. However, it's essential to acknowledge that there exists a threshold beyond which further increments in K yield marginal gains [45]. In this study, we conducted experiments with K values 300 but observed no significant improvement. Consequently, we decided to employ K = 300 for the RF ensemble model architecture, as shown in Fig. 11.

2.4.2. Multi-layer perceptron artificial neural network (MLP-ANN)

Artificial neural networks (ANNs) are inspired by the functioning of biological neurons in organisms' neural networks, offering a unique approach to solving problems that defy traditional mathematical equations. This method enables the intricate modeling of complex systems by establishing relationships between input and target data [2]. The multi-layer perceptron artificial neural network (MLP-ANN) is a prominent AIT, widely employed for modeling intricate non-linear processes in weather-related predictions [46,48,49]. The MLP-ANN typically comprises a feed-forward architecture encompassing an input layer, one or more hidden layers, and an output layer [50]. Training the MLP-ANN commonly involves using the backpropagation algorithm [46]. Activation functions such as linear, hyperbolic tan (sigmoid), and logistic functions are frequently employed to assess their impact on the multi-layer perceptron model [51]. The basic architecture of MLP-ANN

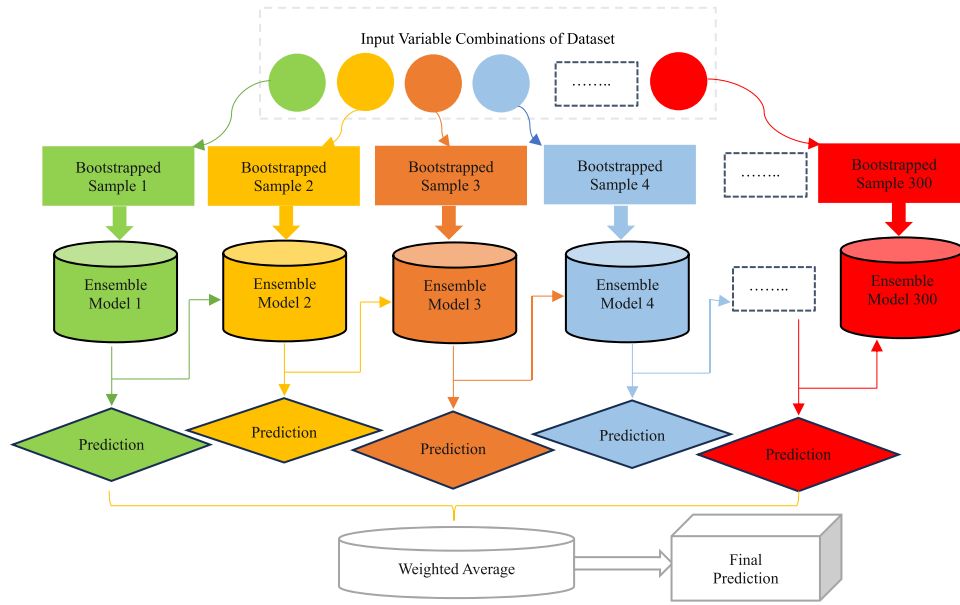


Fig. 11. The basic flow diagram of random forest with bootstrapping aggregation.

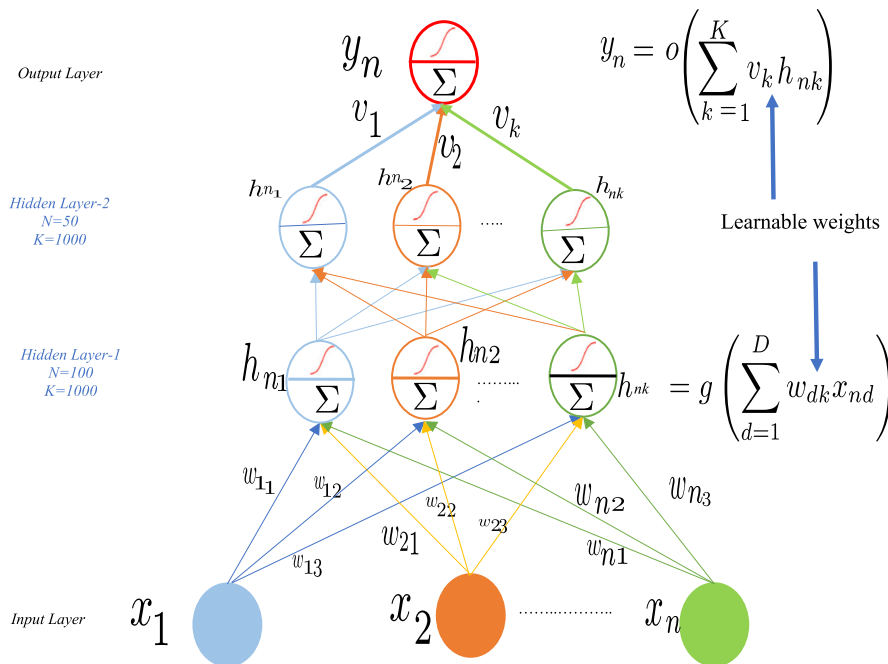


Fig. 12. Working process diagram of MLP-ANN.

is presented in Fig. 12. In developing the MLP-ANN model, an extensive hyperparameter tuning process aimed at minimizing the mean squared error (MSE) metric was shown in Table 5. This process explored various configurations, including hidden layer sizes, activation functions ('relu' and 'tanh'), regularization parameter values (0.0001, 0.001, 0.05), learning rate strategies ('constant' and 'adaptive'), and a maximum of 1000 training iterations. The optimal MLP-ANN configuration, achieved through meticulous tuning, featured an MLP-ANN model with an alpha value of 0.05 and a hidden layer architecture comprising 100 and 50 neurons. With a maximum iteration limit of 1000, this finely tuned model demonstrates superior predictive performance, aligning with the stringent demands of the MSE metric. This rigorous optimization process underscores the model's efficacy for accurate modeling and prediction tasks while adhering to specified hyperparameter constraints.

In this study, the MLP-ANN architecture, the " y_n " represents the output of the model, " k " signifies several iterations, " w " and " v " denotes a neuron's learnable weights at hidden layer 1 and 2, respectively, and " x " symbolizes the input data. Notably, all neurons and layers are assigned weights through training algorithms, making the MLP-ANN a powerful tool for modeling complex relationships between input and target data.

2.4.3. Long short-term memory recurrent neural networks (LSTM-RNN)

The background of LSTM-RNN has already been discussed previously. For the constitution of the prediction monthly rainfall model, the general architecture of the memory cell is presented in Fig. 13. The fundamental concept of LSTM revolves around incorporating an internal state and three essential filter gates within each LSTM cell, denoted as the 'forget gate,' 'input gate,' and 'output gate.' These gates control the

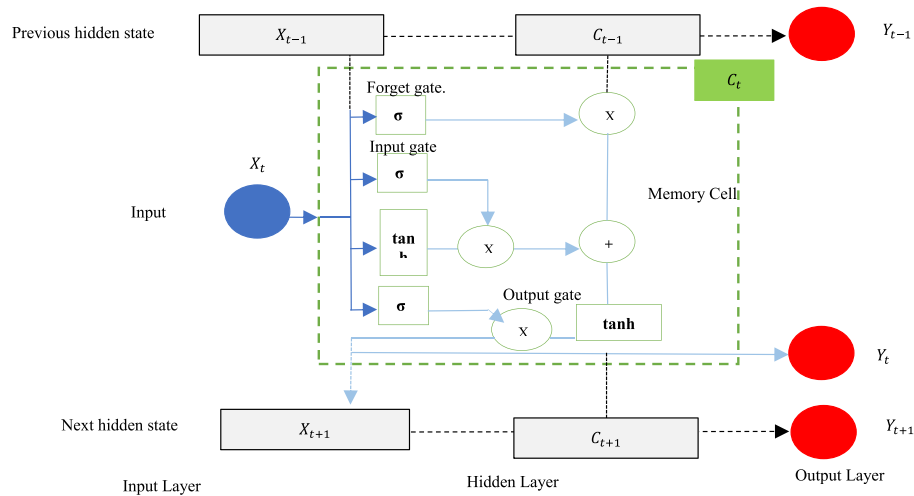


Fig. 13. LSTM-RNN architecture for a working model.

flow of information within the cell at each time step 't,' with inputs derived from 'xt' and the previous hidden state 'ht-1.' The activation functions commonly employed in neural networks and deep learning are sigmoid, tanh, and ReLU (Rectified Linear Unit). In this study, sigmoid and tanh functions are employed. The 'forget gate' employs a Sigmoid function to determine whether communication from the previous time step should be retained or discarded based on its proximity to 1 or 0. The 'input gate' plays a pivotal role in updating communication to the cell state by combining the output of a Sigmoid activation layer with a hyperbolic tangent (tanh) activation layer, resulting in a decision on the state updates within the cell state. The 'output gate' calculates the hidden state value for the next time step by utilizing information from both the 'forget' and 'input gates.' This newly computed hidden state value is essentially the prediction value for the LSTM network.

$$\text{Input Gate} = \sigma(W_i[x_t, h_{t-1}] + b_i) \quad (3)$$

$$\text{Forget Gate} = \sigma(W_f[x_t, h_{t-1}] + b_f) \quad (4)$$

$$\text{Cell candidate} = \tanh(W_c[x_t, h_{t-1}] + b_c) \quad (5)$$

$$\text{Cell State} = C_t = f_t * [C_{t-1} + i_t] * \bar{C}_t \quad (6)$$

$$\text{Output Gate} = (W_o[x_t, h_{t-1}] + b_o) \quad (7)$$

$$\text{Hidden State} = o_t * \text{Sigmoid}(C_t) \quad (8)$$

'W' and 'b' represent the weight matrices and bias at the input, forget cell candidate and output gates. 'X_t' signifies the input to the memory cell, 'ht-1' denotes the previous hidden state at time 't - 1,' and 'C_{t-1}' and 'C_t' refer to the cell states at times 't - 1' and 't'.

Achieving optimal generalization necessitates the calibration of key hyperparameters. These hyperparameters encompass crucial aspects

such as the number of layers, nodes, batch size, verbosity level, and training epochs. The process involves meticulously searching for the most appropriate hyperparameter configurations to enhance the model's predictive capabilities. Therefore, hyperparameter tuning is presented in Table 5. The final model was developed after selecting the best parameters in Table 6. The model produces a single output, which serves as its prediction for monthly rainfall. This output emerges due to the LSTM network's computations based on the input features. The model employs ten dense layers to facilitate this process, introducing non-linearity and complexity into the network. These layers are instrumental in enabling the model to grasp and represent intricate patterns and relationships inherent in the data. The model adopts two distinct activation functions: 'tanh' (hyperbolic tangent) and 'sigmoid.' The Adam optimizer is employed to optimize the model's performance and training process due to its capacity to adapt the learning rate on a per-parameter basis and its effectiveness in handling diverse data types. In terms of training duration, the model undergoes training for 800 epochs, with each epoch representing a complete pass through the entire training dataset. Multiple epochs enable the model to iteratively refine its predictions over time iteratively, gradually improving its forecasting accuracy. Lastly, the primary evaluation metric used to assess the model's performance is MSE. MSE provides an insightful measurement of prediction accuracy by quantifying the average squared difference between predicted and actual rainfall values. This meticulous combination of architecture, hyperparameters, and evaluation metrics underscores the thoughtfulness and expertise behind developing the LSTM RNN model, ensuring its effectiveness in delivering accurate monthly rainfall forecasts.

2.5. Evaluation metrics

This study employs the R², RRMSE, MAE, Pearson's correlation (r), and MAPE statistical evaluation metrics to evaluate the model's suitability. These four statistical parameters' formulae are as follows:

$$R^2 = 1 - \frac{\sum (R_{obs} - R_{pre})^2}{\sum (R_{obs} - R_{avg})^2} \quad (9)$$

$$RRMSE = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (\bar{R}_{Obs}^i - \bar{R}_{pre}^i)^2}}{\sum_{i=1}^n \bar{R}_{Obs}^i} \quad (10)$$

$$MAE = \frac{\sum (R_{pre} - R_{obs})}{N} \quad (11)$$

Table 6
LSTM-RNN components for the final prediction model.

No. of inputs	9
No. of hidden nodes	40
No. of outputs	1
Learning rate	0.01
Dense layers	10
Activation function	['tanh', 'sigmoid']
Optimizer	Adam
Verbose	1
Batch Size	32
Epoch	800
Evaluation Metrics	MSE

$$r = \frac{n(\sum R_{obs} * R_{pre}) - (\sum R_{obs})(\sum R_{pre})}{\sqrt{[n\sum R_{obs}^2 - (\sum R_{obs})^2][n\sum R_{pre}^2 - (\sum R_{pre})^2]}} \quad (12)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n |R_{obs} - R_{pre}| * 100 \quad (13)$$

The R^2 value ranges from 0 to 1 for an effective relationship between predicted and observed values. The model is considered the most effective when the R^2 value is close to or equal to one. A Pearson's correlation (r) might be between -1.00 and $+1.00$. A score of -1.00 represents a perfect negative correlation, whereas a score of $+1.00$ represents a perfect positive correlation, and 0.00 shows no relationship exists between the variables under examination [52]. RRMSE is calculated by dividing RMSE by the average value of measured data. According to [53], model accuracy is considered excellent when $RRMSE < 10\%$, good if $10\% < RRMSE < 20\%$, fair if $20\% < RRMSE < 30\%$, and poor if $RRMSE > 30\%$. The MAE is the sum of the absolute values of the errors divided by the number of observations. This quantity is often used in statistics to measure how close calculated values are to measured values [53]. The MAPE is used to measure the accuracy of a predictive model, particularly in forecasting and regression problems. MAPE typically ranges from 0% (perfect forecast) to 100%+ (significant forecast error), with lower values indicating better forecast accuracy.

3. Results and discussion

3.1. Results

The statistical evaluation of IVS models in the Wang and Nan River basins provides valuable insights into their performance in selection of input variables. The models, including GB, SVR-RFE, and the proposed hybrid model BTSP-LSTM-RNN, were assessed based on MAE, R^2 , RRMSE, (r), and MAPE. Fig. 14 shows the performance of IVS models in the Wang River basin. The R^2 values provide insights into the goodness of fit of these models. GB and SVR-RFE demonstrate R^2 of 0.73 and 0.69, respectively, indicating their ability to explain approximately 73% and 69% of the variance in the target variable. However, BTSP-LSTM-RNN significantly outperforms both with an impressive R^2 value of 0.84,

signifying its capability to capture a higher portion of the complex hydrological patterns in the dataset. GB and SVR-RFE exhibit MAE of 115.56 and 119.50, respectively, implying lower accuracy in predicting river basin responses. In contrast, BTSP-LSTM-RNN attains the lowest MAE of 92.28, highlighting its superior accuracy and suggesting that it provides more precise predictions. The RRMSE further assesses the goodness of fit while considering the variability in the dataset. GB and SVR-RFE display RRMSE values of 13.11 and 13.87, indicating notable variability in their predictions. In contrast, BTSP-LSTM-RNN records a substantially lower RRMSE of 10.36, implying better overall fit and reduced variability. Lastly, the MAPE gauges the relative percentage error in predictions. GB and SVR-RFE models exhibit MAPE values exceeding 100%, suggesting their predictions have a substantial percentage error. While BTSP-LSTM-RNN achieves the lowest MAPE of 29.82%, indicating the least relative percentage error among the models.

In Fig. 15, at the Nan River basin, among three IVS models, the BTSP-LSTM-RNN model demonstrates the highest R^2 of 0.83. It indicates that the model explains a substantial portion of the variance in the target variable, suggesting its capability to capture complex temporal patterns in the hydrological data. Furthermore, it exhibits the lowest MAE of 242.60, implying better accuracy in predicting river basin responses. The RRMSE value of 9.93% for BTSP-LSTM-RNN signifies the goodness of fit of the model, with a focus on the variability in the dataset. It is considerably lower than the other techniques, indicating superior predictive performance. However, it's important to note that the MAPE for BTSP-LSTM-RNN is 30.05%, which suggests that the model's predictions have a moderate percentage error. In contrast, GB and SVR-RFE exhibit slightly lower R^2 of 0.76 and 0.70, respectively. It suggests that these models capture a smaller portion of the variance in the data compared to BTSP-LSTM-RNN. Additionally, both GB and SVR-RFE have higher MAE of 245.97 and 252.61, indicating relatively less accurate predictions than BTSP-LSTM-RNN. The RRMSE values for GB and SVR-RFE are higher at 11.73 and 13.09, respectively, suggesting that these models have a higher level of variability in their predictions.

The statistical evaluation of these IVS models highlights the superior performance of the proposed BTSP-LSTM-RNN hybrid model in terms of R^2 , MAE, RRMSE, and MAPE. It showcases its potential to provide accurate and robust predictions for hydrological modeling in the Wang

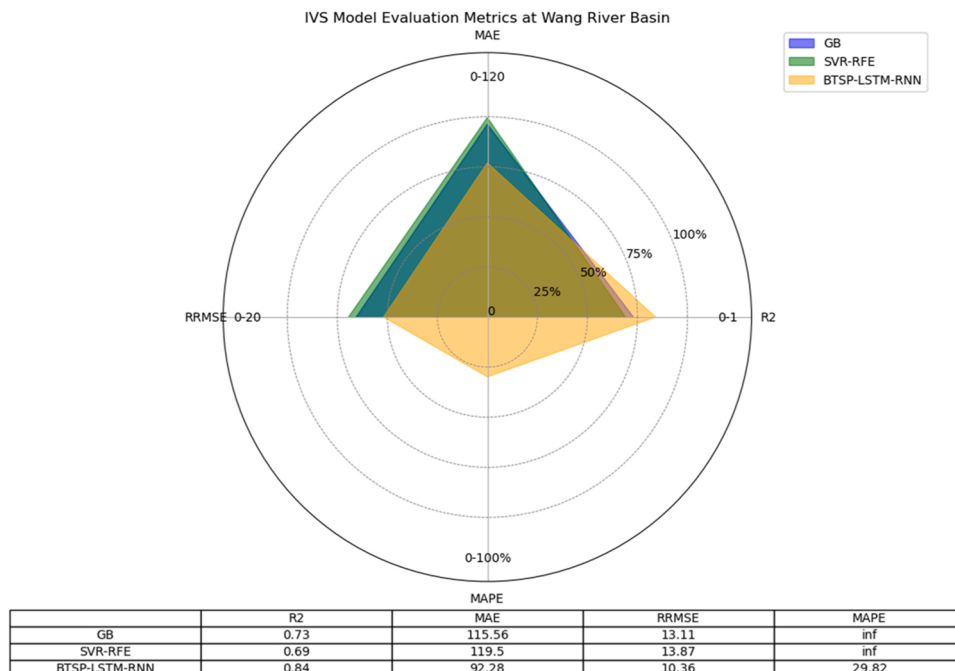


Fig. 14. Statistical evaluation of developed IVS models at Wang River basin.

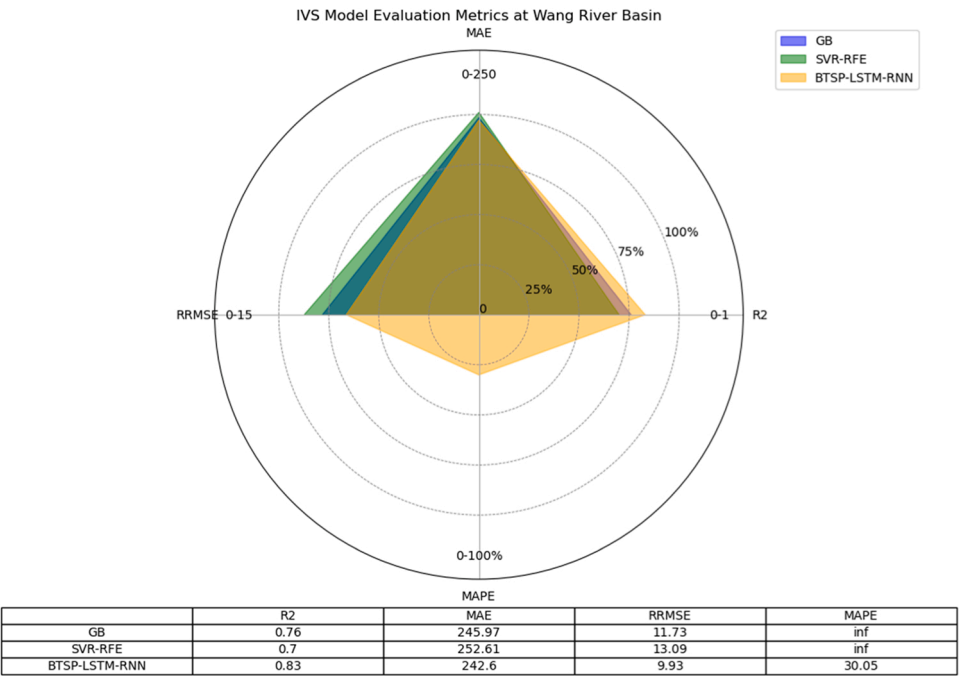


Fig. 15. Statistical evaluation of developed IVS models at the Nan River basin.

and River basin.

Following a thorough statistical evaluation in both the Wang and Nan River basins, the proposed IVS model, BTSP-LSTM-RNN, emerged as the top-performing model. The results of the IVS analysis conducted by the BTSP-LSTM-RNN model underscore its exceptional performance and critical role in IVS for rainfall forecasting. The model’s ability to capture complex hydrological patterns and superior predictive accuracy in both river basins highlight its significance in effectively leveraging various meteorological and climatic features for informed and precise rainfall forecasts. Fig. 16 shows the important input variables at the Wang River basin. RH stands out with an importance score of 3.11. This signifies that RH has the most substantial impact on the model’s ability to predict rainfall, emphasizing the dominant influence of humidity levels on rainfall patterns. Tmin follows closely with an importance score of 1.37, underscoring its crucial role in capturing temperature-related effects on rainfall. Other variables contributing to the model’s predictive power include SST indices, such as Nino 1.2 and Nino 3.4, with importance scores of 0.35 and 0.03, respectively. These indices are

pivotal in understanding larger climatic patterns, such as El Niño and La Niña events, which can profoundly affect regional weather and rainfall. SR also exhibits substantial importance with a score of 0.13, highlighting its role in capturing energy inputs that influence local climate dynamics and rainfall patterns. Additionally, the SOI and WS exhibit importance scores of 0.08 and 0.06, respectively, signifying their contributions to the model’s understanding of meteorological conditions. These factors substantially contribute to the model’s ability to meteorological conditions; others, including WD, Tmax, Nino 4, and the DMI, display relatively lower significance and are eliminated for final selection.

In Fig. 17, for the Nan River basin, BTSP-LSTM-RNN identifies key input variables in line with findings from the Wang River basin. RH, with an importance score of 1.73, stands out, underscoring its substantial impact on rainfall prediction. Tmin, scoring 1.4337, plays a crucial role in capturing temperature-related effects on rainfall, particularly night-time conditions affecting condensation and dew point. Additionally, PS and SST indices (Nino 1.2, Nino 3, Nino 3.4) hold significant importance

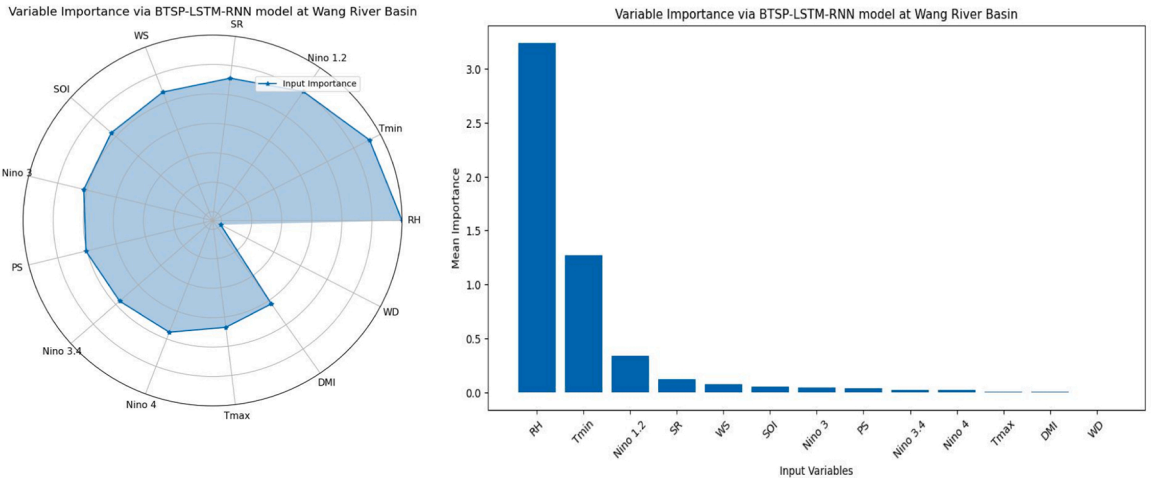


Fig. 16. Input variables importance via BTSP-LSTM-RNN model at Wang River basin.

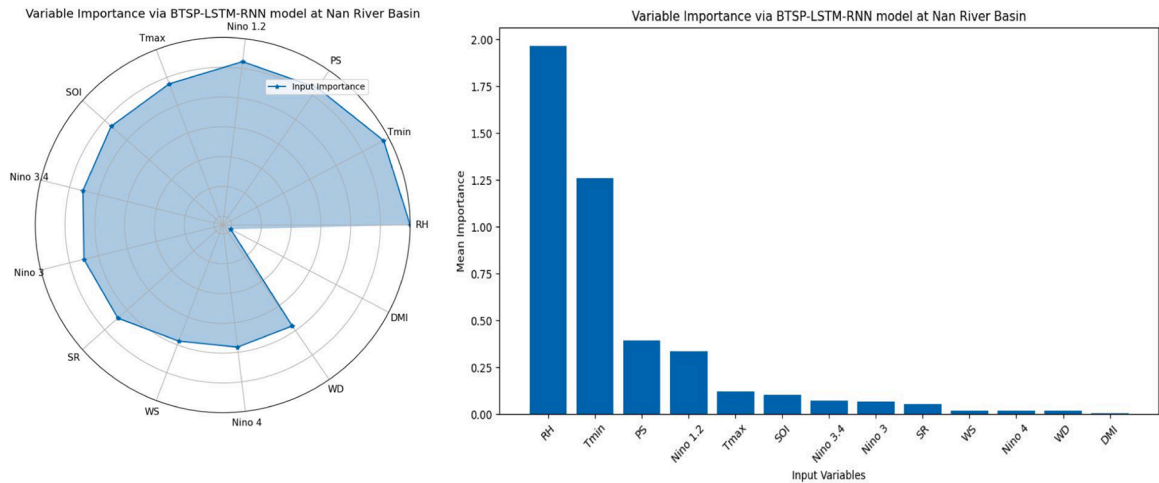


Fig. 17. Input variables importance via BTSP-LSTM-RNN model at Nan River basin.

scores of 0.38, 0.27, 0.08, and 0.05, respectively, aiding in understanding broader climatic patterns, including El Niño and La Niña’s influence on regional weather. Tmax and SR, with scores of 0.11 and 0.05, respectively, contribute to modeling wind dynamics and energy inputs on local climate. Meanwhile, variables like WS, WD, Nino 4, and

DMI, with lower importance scores, are excluded during subsequent input variable combination phases to streamline the model and enhance its precision in capturing meteorological complexities in the Nan River basin.

This analysis aimed to enhance the model’s applicability and

Comparison of Prediction Models Outputs with both Input Combinations (IVC-1 and IVC-2) at Wang River Basin

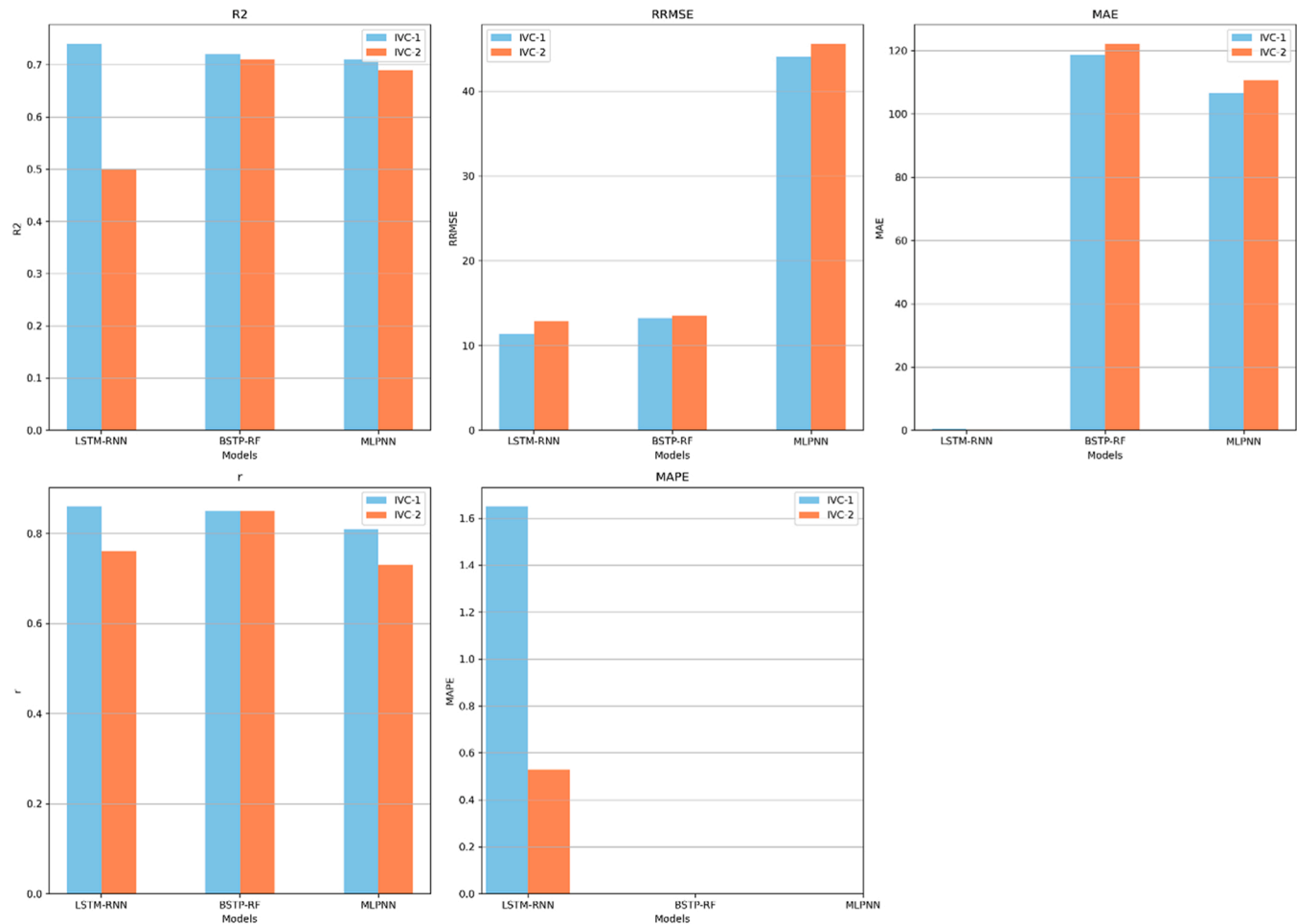


Fig. 18. Comparison of prediction models based on R^2 , RRMSE, r , and MAE at Wang River basin.

accuracy, ultimately contributing to more reliable predictions in rainfall forecasting. Based on IVS model (BTSP-LSTM-RNN) outputs and final selected inputs from overall data, this study is composed of the following two input variable combinations (IVC):

IVC-1: Models with variables selected by BTSP-LSTM-RNN input selection model by eliminating less important variables as input.

IVC-2: Models with all original meteorological and large-scale climatic variables without using the IVS model as inputs.

To predict monthly rainfall at the Wang and Nan River basins, three distinct models (LSTM-RNN, MLPNN, and BSTP-RF) were employed and evaluated based on their performance across six different IVCs as IVC-1 and IVC-2. The evaluation criteria encompassed key hydrological metrics: R^2 , RRMSE, MAE, and Pearson's correlation (r). Fig. 18 presents the results of prediction models at the Wang River basin. In the case of LSTM-RNN, it is observed that the model achieved a commendable performance for both IVC-1 and IVC-2, as indicated by the R^2 of 0.74 and 0.5, respectively. These values suggest that LSTM-RNN captures a substantial portion of the variance in monthly rainfall, particularly for IVC-1. The relatively low RRMSE percentage values of 11.39% and 12.87% further underscore the model's accuracy prediction capability. However, the model's performance regarding MAE and the " r " is more favorable for IVC-1 than IVC-2. This may imply that the choice of IVS plays a crucial role in the model's predictive accuracy. Additionally, the MAPE values of 1.65% and 0.53% for IVC-1 and IVC-2, respectively, indicate that LSTM-RNN provides reliable predictions with minimal relative error. Turning to the BSTP-RF model, it is evident that this approach also

demonstrates promising predictive capabilities. IVC-1 and IVC-2 exhibit relatively high R^2 values of 0.72 and 0.71, respectively, implying that the model captures a substantial portion of rainfall variability. However, it is important to note that the RRMSE percentages for IVC-1 and IVC-2 are notably higher than LSTM-RNN, standing at 13.24% and 13.53%, respectively. Furthermore, the MAE values of 118.62 for IVC-1 and 122.16 for IVC-2 indicate that BSTP-RF tends to produce predictions with higher absolute errors. The " r " of 0.85 for both IVCs highlights the consistency of the model's performance. Unfortunately, the model exhibits infinite MAPE values, suggesting that it occasionally produces predictions close to zero, resulting in an undefined relative error. In the case of MLP-ANN, it is evident that the model's performance is the weakest among the three approaches. The R^2 values of 0.71 for IVC-1 and 0.69 for IVC-2 indicate that the model explains a relatively lower rainfall variability than LSTM-RNN and BSTP-RF. The RRMSE percentages of 44.09% and 45.58% for IVC-1 and IVC-2, respectively, highlight the significantly higher errors in predictions produced by MLP-ANN. The MAE values of 106.56 and 110.66 further emphasize the model's tendency to generate predictions with substantial absolute errors. Moreover, the correlation coefficient (r) values of 0.81 for IVC-1 and 0.73 for IVC-2 suggest weaker associations between predicted and observed values. Like BSTP-RF, MLP-ANN exhibits infinite MAPE values, indicating instances of predictions close to zero.

Overall, the evaluation results suggest that LSTM-RNN outperforms BSTP-RF and MLP-ANN in predicting monthly rainfall for the Wang River basin. Furthermore, the choice of input variables (IVC-1 versus

Comparison of Prediction Models Outputs with both Input Combinations (IVC-1 and IVC-2) at Nan River Basin

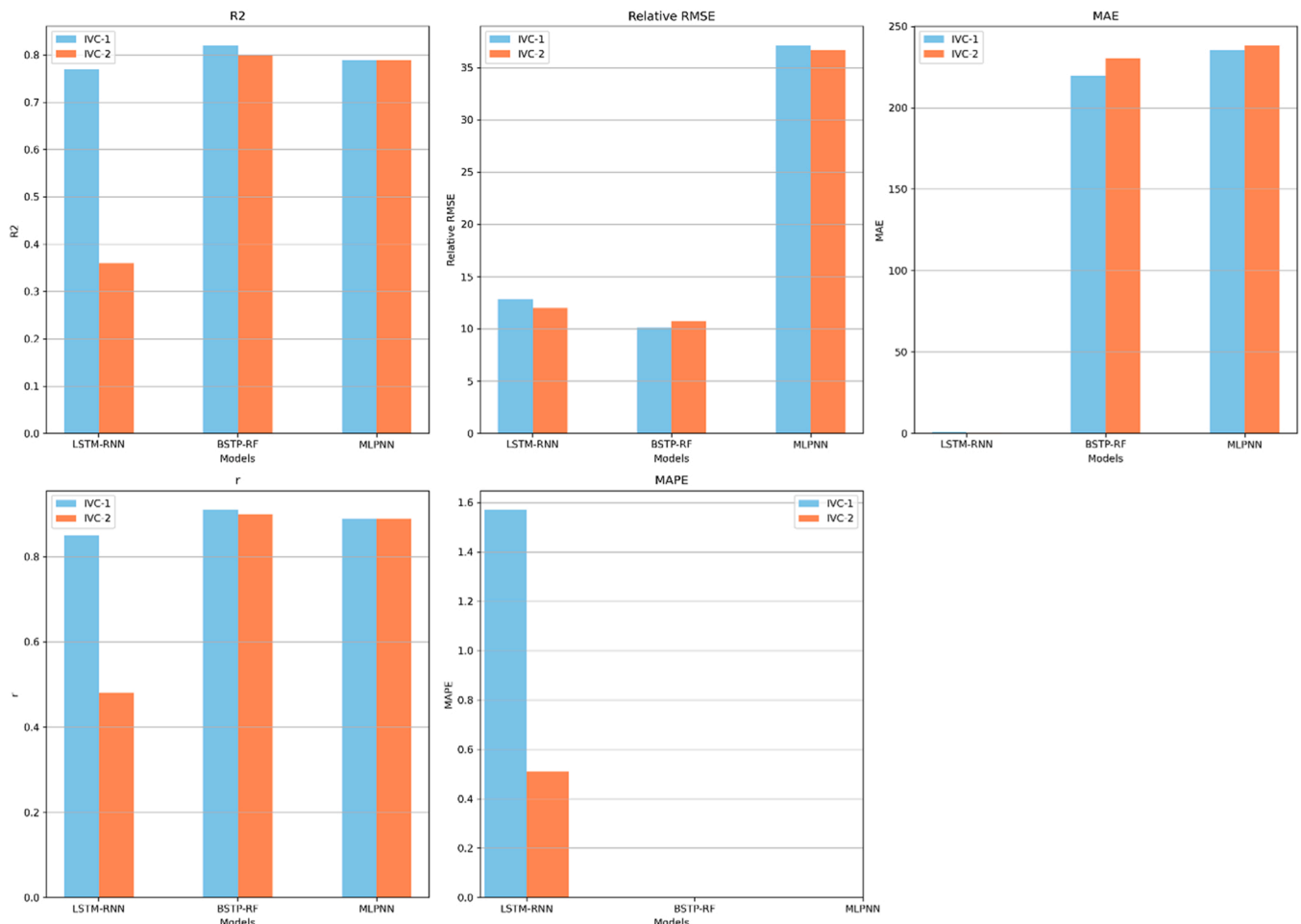


Fig. 19. Comparison of prediction models based on R^2 , RRMSE, r , and MAE at Nan River Basin.

IVC-2) significantly influences the model performance, with IVC-1 generally yielding superior results. Results show that the IVS (novel BSTP-LSTM-RNN hybrid model) selected more relevant input variables for monthly rainfall forecasting. These findings underscore the importance of selecting appropriate input variables in rainfall forecasting in the Wang River basin, with LSTM-RNN demonstrating the potential for accurate and reliable predictions. Nonetheless, further research may be warranted to address the limitations observed in the BSTP-RF and MLP-ANN models.

In the Nan River basin, Fig. 19 depicts the LSTM-RNN model's performance. For IVC-1, the model boasts a substantial R^2 of 0.77, explaining 77% of observed rainfall variance, and a low RRMSE of 12.82%, indicating close alignment with actual values. The MAE of 0.67 demonstrates reasonable accuracy, and a high "r" value (0.85) signifies a strong linear relationship, supported by a low MAPE of 1.57%, highlighting proficiency in capturing rainfall patterns. However, when applied to IVC-2, LSTM-RNN's performance drops significantly. R^2 declines to 0.36, suggesting difficulty in explaining variance. RRMSE remains at 12.00%, while MAE decreases to 0.25, and "r" drops to 0.48, indicating a weaker linear relationship. MAPE decreases to 0.51%, implying improved but still suboptimal performance. Moving on to the BSTP-RF model, it demonstrates robust performance for both IVC-1 and IVC-2. R^2 values are notably high (0.82 and 0.80), indicating strong variance explanation. RRMSE values around 10% demonstrate a good fit with low prediction errors. However, high MAE values (219.58 and 230.52) suggest potential bias in predictions. The high "r" (0.91 and 0.90) supports a strong linear relationship, yet infinite MAPE values signal potential outlier issues, warranting further investigation. The MLP-ANN model exhibits reasonably good performance for both IVCs with an R^2 of 0.79, explaining a significant portion of rainfall variance. However, RRMSE values around 37% indicate higher prediction errors. Relatively high MAE values (235.31 and 238.2) suggest some bias in predictions. The "r" (0.89 for both) demonstrates a strong linear relationship. Like BSTP-RF, infinite MAPE values suggest potential outliers or extreme errors in predictions. Overall, BSTP-RF consistently performs well in both IVCs but requires attention to outliers. In contrast, LSTM-RNN and MLP-ANN exhibit variable performance depending on the IVC, with LSTM-RNN performing better for IVC-1, and both models

showed higher errors for IVC-2 (Figs. 20 and 21).

These findings align well with [20], which indicated that adding the RH, Tmin, WS, and PS can forecast accuracy. As anticipated, performance improved when meteorological variables (T, P, RH, and W) were incorporated into IVC-1. In line with [6], meteorological and LCVs can increase the accuracy of monthly rainfall forecasting. In the context of monthly predictions for both River basins, the IVS model BSTP-LSTM-RNN demonstrated satisfactory performance overall in variable selection and also aligns with the findings of a study by [7,11, 31] which focused on rainfall prediction and in cooperation with selected variables for the prediction models. It's important to note that the impact of weather variables on predicted rainfall varies across lead times and stations [54]. Based on prediction model results shown in Fig. 22, the BTSP-LSTM-RNN (IVS) model plays a pivotal role in the selection of input variables for rainfall forecasting and its impact on the performance of prediction models (BSTP-RF, MLP-ANN, and LSTM-RNN). When BTSP-LSTM-RNN's selected input variables from IVC-1 are utilized, the BSTP-RF model demonstrates robust performance shown in Figs. 20 and 21 for both river basins. It achieves a high R^2 (0.82), indicating its ability to explain a substantial portion of the variance in monthly rainfall. The low RRMSE (10.13%) suggests accurate predictions, and the correlation coefficient (r) of 0.91 further supports the model's strong linear relationship with observed rainfall data. The presence of infinite MAPE values may require further investigation, but overall, IVC-1 derived from BTSP-LSTM-RNN contributes to the BSTP-RF model's success. Like BSTP-RF, the MLP-ANN model benefits from IVC-1. It achieves a high R^2 (0.79) and a relatively low RRMSE (37.10%), indicating the model's ability to explain a significant portion of the variance and produce reasonably accurate predictions. The correlation coefficient (r) of 0.89 further supports its performance. However, like BSTP-RF, the model exhibits infinite MAPE values, which may need further attention. LSTM-RNN results in a commendable R^2 (0.74), relatively low RRMSE (11.39%), and a low MAE. The correlation coefficient (r) of 0.84 highlights the strong linear relationship between model predictions and observed rainfall. In line with Liyew and Melese's research [55], strong correlations are indicated by r values greater than 0.6 but less than 0.8 when interpreting the correlation coefficient range. Additionally, the low MAPE (1.65%) indicates reliable predictions.

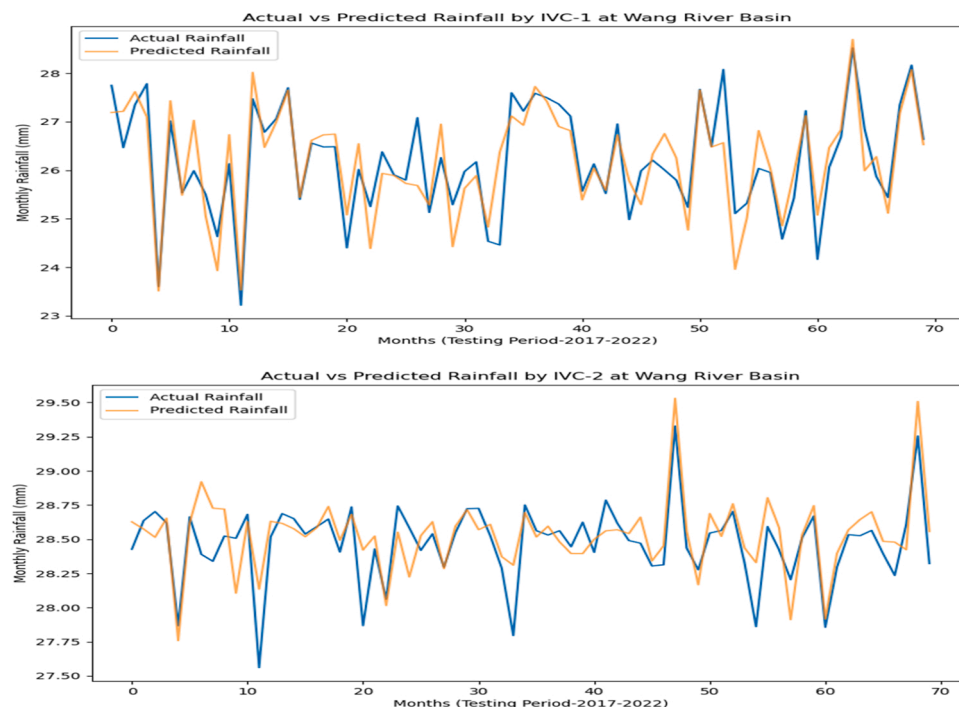


Fig. 20. LSTM-RNN models results for IVC-1 and IVC-2 at Wang River basin.

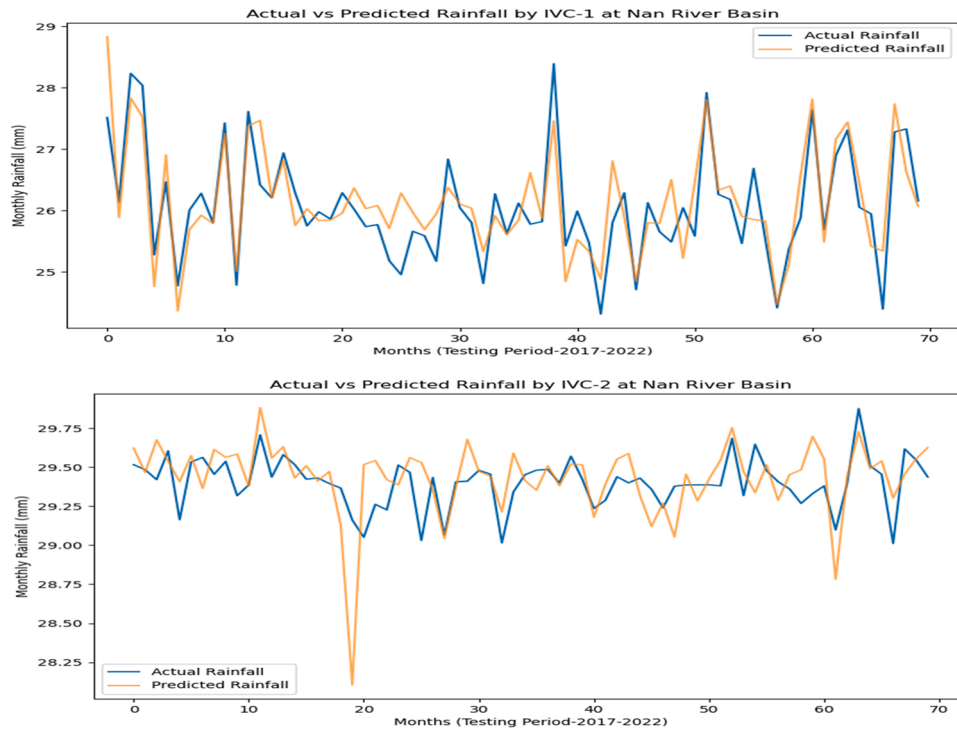


Fig. 21. LSTM-RNN models results for IVC-1 and IVC-2 at Nan River basin.

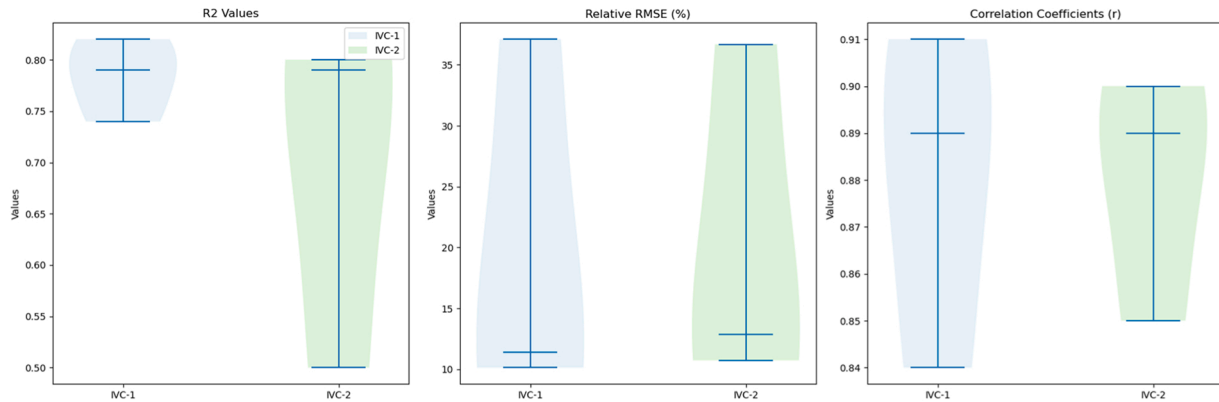


Fig. 22. Overall performance of prediction models with both input variable combinations (IVC-1 and IVC-2).

Overall, BTSP-LSTM-RNN's role in IVS enhances R^2 , reduces RMSE, and strengthens linear relationships for accurate rainfall predictions. While IVC-2 performs reasonably well, IVC-1's BTSP-LSTM-RNN variable selection consistently yields superior results, emphasizing the IVS model's significance in hydrological modeling and rainfall forecasting.

4. Conclusion

This study proposed a hybrid approach, integrating various artificial intelligence techniques for IVS, notably the SVR-RFE, GB, and the novel BTSP-LSTM-RNN. The performance of these hybrid AITs models is evaluated using the R^2 , RMSE, MAPE, Pearson correlation (r), and MAE. After determining the optimal IVS, selected variables are systematically combined into two input variable combinations (IVC-1; the first is based on the IVS model's selected variables, and IVC-2 is used without the IVS model application). After the prediction model development phase, all predictive models undergo a thorough evaluation using performance metrics such as R^2 , RMSE, Pearson correlation (r), and MAE. In the Wang River basin, GB and SVR-RFE exhibited R^2 values of 0.73 and 0.69,

respectively, while BTSP-LSTM-RNN significantly outperformed them with an impressive value of 0.84. Furthermore, BTSP-LSTM-RNN demonstrated the lowest MAE of 92.28, highlighting its superior accuracy. The RRMSE was substantially lower at 10.36 for BTSP-LSTM-RNN, indicating better overall fit and reduced variability. While all models exhibited MAPE values exceeding 100%, BTSP-LSTM-RNN achieved the lowest MAPE of 29.82%. In the Nan River basin, BTSP-LSTM-RNN also emerged as the top-performing model with the highest R^2 value of 0.83. The model showcased excellent accuracy with the lowest MAE of 242.60 and a relatively low RRMSE of 9.93%. However, its MAPE was 30.05%, suggesting a moderate percentage error. The results consistently highlighted the superiority of BTSP-LSTM-RNN in both river basins, emphasizing its potential for accurate and reliable rainfall forecasting. The model's impact on the selection of input variables was crucial, with IVC-1 derived from BTSP-LSTM-RNN consistently yielding superior results compared to IVC-2. The variable selection process significantly enhanced the model's performance, highlighting the importance of the IVS model in hydrological modeling and rainfall forecasting. This study also emphasizes the potential of LSTM-RNN as a robust and accurate

choice for such applications. While BSTP-RF and MLP-ANN models showed promising performance, they exhibited limitations, such as outliers and prediction errors, suggesting areas for further investigation and refinement. However, it's important to note that the performance of these models can be sensitive to the specific characteristics of the river basin, including climate, topography, and data availability. These characteristics should be included for more precise and accurate forecasting for future studies. Also, it is recommended to use this proposed methodology for other components of the hydrological cycle.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

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